

A comparison of Monte-Carlo generated Muon Maps with near horizontal SD showers

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Abstract

For events with zenith angles $> 60^\circ$, the geomagnetic field introduces radial asymmetries to the lateral particle distribution of the shower. A reconstruction in this special case is possible with Monte-Carlo derived maps of muon densities, called *muon maps*.

Currently, three map types generated by different methods and using different interaction models in their simulation are available, including a new type of map presented in this note.

The three maps are compared with each other and with real data. Combining different approaches, we try to find the muon map that provides the best description of the data. Furthermore, an estimate of the systematic uncertainty introduced to the energy reconstruction by using a certain map is given and the influence of the inclined energy spectrum shown.

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1 Introduction

The Surface Detector (SD) of the Pierre Auger Observatory is sensitive to hadronic showers with zenith angles $0^\circ < \theta < 90^\circ$. It consists of an array of individual Water-Cherenkov-Detectors, which sample the extensive particle shower front on the ground. The front consists mainly of photons, electrons and muons (see Fig. 20 in the appendix). The signal of an individual station S_i is roughly proportional to the number of particles which fall into the tank N_i^{tank} .

$$S_i \propto N_i^{\text{tank}} \text{ (approx.)} \quad (1)$$

The approximation is very good for muons as their signal is energy independent to a high degree. It is not so good for photons or electrons, as they produce subshowers inside the water volume of the tank, which enhance the signal depending on the particle energy.

To reconstruct the primary energy E from this information, we can exploit, that the relation of E to the total particle number on the ground at a fixed zenith angle $N^{\text{ground}}(\theta)$ can be approximated very well by a power law

$$N^{\text{ground}}(\theta) \propto E^{1/\gamma} \Leftrightarrow E = E_c \cdot N^{\text{ground}}(\theta)^\gamma \quad (2)$$

whereas E_c and γ are approximately constant within the limits of the Auger energy range and in case of a constant primary composition [1]. Semi-empirical calculations agree with data and simulations, that $\gamma \gtrsim 1$ for a hadronic cascade. The particle number on the ground N^{ground} is a function of θ , because the atmospheric attenuation increases with growing zenith angle.

For $0^\circ < \theta < 60^\circ$ a standard reconstruction method for E exists. It uses $S_{1000}(\theta)$ as an energy estimator, which is the interpolated tank signal in a lateral distance of 1000 m to the shower axis at a fixed zenith angle. This exploits, that the lateral signal profile of the shower has a radial symmetry and a shape, that is approximately independent of E .

$$S_{1000}(\theta) \propto N^{\text{ground}}(\theta) \propto E^{1/\gamma} \quad (3)$$

The θ -dependency can be factorised and derived from data by a constant intensity cut (CIC) method, but this is not the scope of this note.

Reconstructing the energy of inclined showers with $\theta > 60^\circ$ imposes two additional challenges.

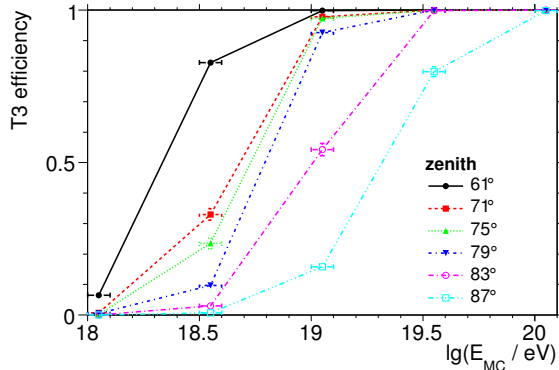


Figure 1: Simulated efficiency of the T3 for $\theta > 60^\circ$. The plot was generated from 18000 simulated SD events using a CORSIKA/QGSJet-II/FLUKA library of proton showers [3–6] and the detector simulation of Offline-v2.0 [7]. For the parameter distribution of the library see Table 4.

First, the signal on the ground is stronger attenuated and spread wider by the longer path through the atmosphere. This affects the threshold energy $E_{\text{thres}}^{\text{T3}}$, above which the SD event trigger chain is saturated^{a)}. Second, the signal distribution lateral to the shower axis is not radially symmetric anymore, because of geomagnetic distortions, geometric effects and different attenuations of early and late arriving particles (see Fig. 20 in the appendix).

At the same time, detecting inclined showers also provides valuable advantages. The two main benefits are increased event statistics and a larger field of view of the sky. Simulations show, that the Auger event trigger system allows to enlarge the observable zenith angle range up to $\theta = 80^\circ$ for shower energies at least above $E_{\text{thres}}^{\text{T3}} \approx 10^{19}$ eV (see Fig. 1 and [2]). The zenith angle range $\theta \in [60^\circ, 80^\circ]$ contains 22 % of all SD events.

There are other advantages as well. At $\theta = 70^\circ$, the primary electromagnetic cascade of the shower is totally absorbed, which allows the isolated observation of the muon signal on the ground. Also, understanding inclined showers means understanding the main background to neutrino searches.

Because of the asymmetries present in inclined showers, the inclined energy reconstruction follows a different idea to estimate $N^{\text{ground}}(\theta)$. The mea-

^{a)} Trigger saturation is necessary for a reliable exposure calculation.

sured tank signals are compared to Monte-Carlo derived muon density profiles, called *muon maps*. Muon maps are basically a 2-dimensional parameterisation of the average muon number per tank

$$\bar{N}_\mu^{\text{tank}} = \bar{N}_\mu^{\text{tank}}(x, y; \theta, \phi) \quad (4)$$

for each position (x, y) in the lateral shower plane at a particular combination of zenith angle θ and azimuth angle ϕ . They are generated at a fixed energy of 10^{19} eV and primary mass (usually protons) and try to model all asymmetries as well as the total attenuation with zenith angle.

The overall scaling factor, which gives the best fit between the estimated signals from a muon map $\bar{N}_\mu^{\text{tank}}(x, y; \theta, \phi)$ and a measured SD event, is called N_{19} . N_{19} is related to the primary energy:

$$N_{19} \propto N^{\text{ground}}(\theta) \propto E^{1/\gamma} \Leftrightarrow E = E_{\text{MC}} \cdot N_{19}^\gamma \quad (5)$$

The θ -dependency here is absorbed directly into the muon map.

Unfortunately, E_{MC} and γ calculated from current simulations are inconsistent with data [8]. Instead, these constants are therefore derived from a fit of coincidentally measured events in the Surface and Fluorescence Detector (hybrid calibration). The FD energy E_{FD} is a good estimator for the true energy of the shower E , but in reality, of course, it has its own uncertainties and systematics [9].

Because the muon map $\bar{N}_\mu^{\text{tank}}(x, y; \theta, \phi)$ is currently generated from shower simulations, it is sensitive to the choice of the implemented interaction models, as well as the method to convert the Monte-Carlo data into the final muon map. A muon map that does not reproduce the muon density profile observed in the data might introduce systematic shifts to the reconstructed energy and decrease the energy resolution. Two different types of maps currently exist [10, 11] with the addition of a new one, first presented in this note.

The new type of map was generated from a CORSIKA proton shower library [3]. In contrast to the other two map types which used the AIRES/QGSJet01 [12, 13] combination for the shower simulation, CORSIKA/QGSJet-II [4, 5] was used for the new map. Another important difference is, that the full simulation of the shower up to the ground was done within CORSIKA. The information of all shower components on the ground is

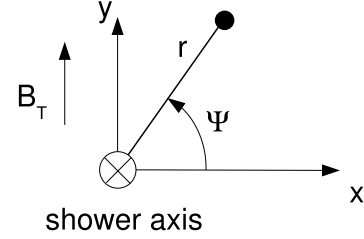


Figure 2: Shower plane coordinate system. e_z is anti-parallel to the momentum vector of the shower, e_y is parallel to B_T , the component of the geomagnetic field perpendicular to the shower axis. ψ and r are polar coordinates in this lateral plane.

available, including the distribution of the electromagnetic component. A fourth muon map, based on the EPOS interaction model [14] is in preparation.

The purpose of this note is to compare in detail the three different muon maps, to show possible systematic effects on the reconstructed energy and the inclined energy spectrum by choosing one or another, as well as to inspect the agreement with signal profiles found in real data.

2 Coordinate system

For analysing inclined showers, it is very useful to project quantities measured on the ground into a special plane shown in Fig. 2. This plane is perpendicular to the shower axis and orientated so, that the projected magnetic field is parallel to the y-axis. We will use this coordinate system in the whole note.

Regarding the shower development, we will also distinguish between two important distances. The first is the vertical height of the shower maximum h_{max} , the other is the path length between the shower maximum and the arrival point on the ground d_{max} (see Fig. 3). The shower maximum will always be defined as the point along the shower track, where the *electromagnetic* particles reach their maximum number, the production point of most muons however is very close to this point.

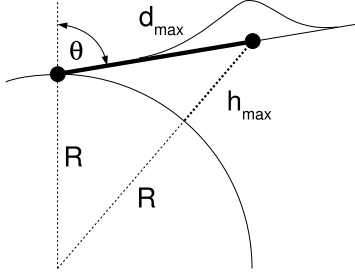


Figure 3: Scheme for defining the vertical altitude of the shower maximum above sea level $h_{\max}$, and its distance from the surface along the shower path $d_{\max}$. R is the radius of Earth, θ the inclination at the shower arrival point.

The two distances are related:

$$d_{\max}(\theta, h_{\max}) = R \left(\sqrt{\left(1 + \frac{h_{\max}}{R}\right)^2 - 1} + \cos \theta \right) \quad (6)$$

with R being Earth's radius.

Beware: In this note, we will always express $h_{\max}$ and $d_{\max}$ with respect to sea level, *not* with respect to the surface level in Malargüe.

3 Muon Maps

We will start with a general overview about the signal distribution of inclined showers on the ground and explain the ansatz of muon maps to model these signals. We will show an overview of the currently available muon maps, made by different groups. Finally, a new map will be presented, which was generated using an improved technique from a CORSIKA proton shower library.

3.1 Surface detector signals

The muon signal in a Cherenkov-detector is in very good approximation independent of the muon energy. Electrons and photons on the other hand produce an energy dependent signal, since these particles can produce subshowers in the water volume if they are highly energetic enough and thus multiply the amount of produced Cherenkov light.

Which signal dominates in the surface detector tank is basically a function of the amount of absorption in the atmosphere and distance to the shower axis. Since muons are 200 times heavier than electrons, they only loose a small amount of their initial energy by scattering on the way to the ground. Electrons and photons on the other hand are only short lived and get absorbed by the atmosphere as soon as the hadronic cascade runs out of energy.

To get an overview over the amount of transversed matter density and the involved path lengths from the shower maximum to the ground, we calculate the integrated atmospheric depth X_{atm} , distance $d_{\max}$ and height of the shower maximum $h_{\max}$ numerically, using the parameterisation of the US-Standard Atmosphere in [15] and a two-line-fit of the depth of shower maximum from Auger data in [16]

$$\begin{aligned} X_{\max} &= 726 \pm 1 \text{ gcm}^{-2} + \\ &\quad c_i \times (\lg(E/\text{eV}) - 18.45 \pm 0.05) \\ c_1 &= 80 \pm 3 \text{ gcm}^{-2} \text{ for } \lg(E/\text{eV}) \leq 18.45 \\ c_2 &= 35 \pm 4 \text{ gcm}^{-2} \text{ for } \lg(E/\text{eV}) > 18.45 \end{aligned} \quad (7)$$

The result is shown in Fig. 4. The important point is, that above $\theta = 60^\circ$ the integrated atmospheric depth X_{atm} along the shower axis increases dramatically. The same is true for the distance $d_{\max}$ between the shower maximum and the arrival point on the surface. The vertical height $h_{\max}$ increases more slowly, but also by an order of magnitude if we go from $\theta = 0^\circ$ to 90° .

With these plots in mind, we can understand the tank signal composition in SD events. The electromagnetic signal fraction in Auger surface tanks is seriously reduced in inclined showers (see Fig. 5). Between 60° and 70° , the primary electromagnetic component of the shower generated by π_0 decays becomes extinct and the average signal fraction drops down. The small remaining electromagnetic signal is generated by the muon themselves, mostly by decays and rarely by nuclear interaction. Above $E_\mu \approx 2 \text{ TeV}$, also direct pair production and bremsstrahlung contribute [4]. The resulting electromagnetic distribution necessarily follows the muon distribution closely.

Above $\theta = 80^\circ$ and near the core, the electromagnetic fraction is larger, because the mean muon energy $\bar{E}_\mu$ is higher. We can understand this by the

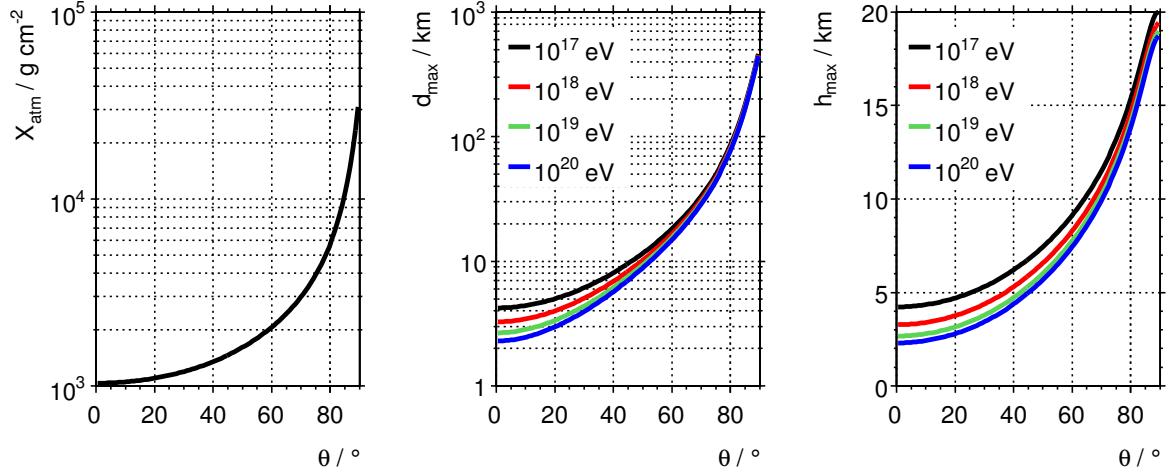


Figure 4: From left to right vs. zenith angle θ : 1) integrated atmospheric depth X_{atm} , 2) distance between shower maximum and arrival point on the surface d_{max} and 3) vertical height of the shower maximum h_{max} . For the calculation of d_{max} and h_{max} , we used the parameterisation of the US-Standard-Atmosphere and the measured depth of the shower maximum from Auger data, as described in the text.

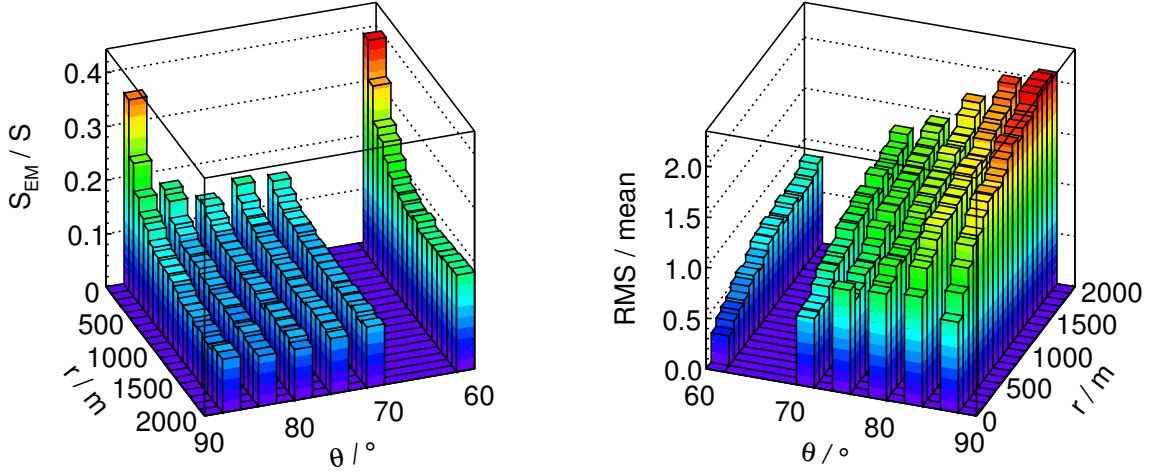


Figure 5: Simulated fraction of the electromagnetic tank signal in inclined showers: (left) the mean and (right) the fluctuation, using the same set of events as in Fig. 1. Shower energies between $E = 10^{18}$ eV and $E = 10^{20.1}$ eV are included (see Table 4). The average fraction has only a weak dependency on primary energy E , while the fluctuation of S_{em}/S decreases with the energy from up to 400 % around 10^{18} eV to up to 200 % around 10^{20} eV. The fluctuation increases in regions with low signals, where it is very likely, that a tank will be hit by only muons or only elm. particles. The fluctuation might be overestimated to some extent, because of artificial fluctuations from the thinning of elm. particles and muons in the simulation. The influence of artificial fluctuations needs to be investigated in a future study.

Lorentz boost, which concentrates muons with high energies near the axis. The increase of $\bar{E}_\mu$ at ground with zenith angle θ (see Fig. 20 in the appendix) is caused by two other effects. The first point is, that the altitude of the production point $\approx h_{\max}$ increases with θ and thus the atmospheric density at the shower maximum ρ_{atm} decreases. Following this trend, charged Pions and Kaons from the shower core are more likely to decay into muons directly, before making another nuclear interaction which would further dissipate their energy. The second point is, that the increasing atmospheric depth absorbs more muons of low energy. Thus the average muon energy is shifted upwards.

Still in most cases, the tank signal on the ground for nearly horizontal showers is dominated by muons. The current commonly used reconstruction ansatz is therefore to use the mean muon number per tank from the muon map $\bar{N}_\mu^{\text{tank}}(x, y; \theta, \phi)$ to generate a probability density function (pdf) for the expected muonic tank signal for each tank $p(S_{\mu,i})$. Included into the pdf is a Poisson distribution for the muon number, convoluted with the signal pdf of a single muon hitting the tank.

The predicted signal distribution $p(S_{\mu,i})$ is then compared in a log-likelihood-minimisation to the measured signal S_i , after correcting the signal for an average electromagnetic signal contribution $\bar{f}_{\text{elm}}(r, \theta, E)$ [11, 17]:

$$S_{\mu,i}^{\text{corr}} = S_i(1 - \bar{f}_{\text{elm}}(r, \theta, E)) \quad (8)$$

with $f_{\text{elm}} = S_{\text{em}}/S$. The reconstruction code *efit* of the USC Santiago de Compostela group [17] used later in this node follows this ansatz.

While this ansatz works for the average signal, the predicted signal fluctuation will be underestimated, since the fluctuation of f_{elm} is not considered. The fluctuation of f_{elm} is large and its distribution non-trivial. It needs to be investigated further in the future.

3.2 Geomagnetic deformation

The electromagnetic component and the muonic component in a shower behave quite differently. Electromagnetic particles travel only short distances but are continuously produced from the hadronic shower core. In contrast, most muons are created in a rather confined region near the shower

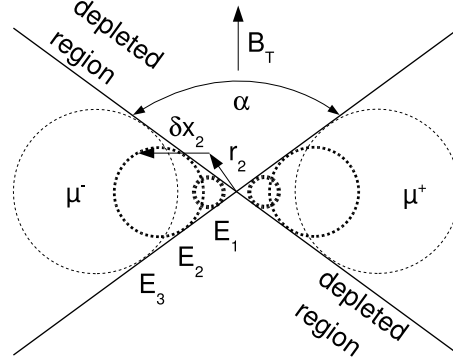


Figure 6: Schematic view of the geomagnetic field effect on the lateral muon distribution. With the magnetic field turned off, relativistic muons with equal energies tend to gather in circles around the shower axis: $r_i \propto E_{\mu,i}^{-1}$ with $E_{\mu,1} > E_{\mu,2} > E_{\mu,3}$. With the magnetic field turned on, these circles get a shift along e_x which is proportional to the same power in energy: $\delta x_i \propto E_{\mu,i}^{-1}$ (see text). α is the angular size of the depleted region, which forms for $\delta x \geq r$ [10].

maximum and after that propagate in near straight lines to the ground, only weakly affected by multiple scattering, ionisation energy loss and decay.

At the Auger South site, currently a geomagnetic field of $24.5 \mu\text{T}$ with zenith angle $\theta_{\text{mag}} = 55^\circ$ and azimuth angle $\phi_{\text{mag}} = 87^\circ$ can be observed on the ground [18]. The field is slowly changing with time. After the 20 year life time of Auger, the strength is expected to decrease by $\approx 5 \%$ and the direction to change by 1° to 2° .

Without the magnetic field, muons would propagate in near straight lines to the ground. Upon neglecting multiple scattering and ionisation losses, relativistic muons with $E_\mu \simeq p_\mu c$ have a simple relation between energy E_μ , distance from the production point d and radial distance from the shower axis r [10]:

$$r \simeq \frac{p_T^\mu}{p_\mu} d \simeq \frac{c p_T^\mu}{E_\mu} d \quad (9)$$

p_T is the transverse momentum of the muon.

With the geomagnetic field in place, the component perpendicular to the shower axis B_T separates positive and negative muons. The magnetic deflec-

$\theta/^\circ$	$d_{\max}/\text{km}$	$a_{\max}$	$\alpha_{\max}/^\circ$
0	2.7	0.02	-
60	16	0.2	-
70	30	0.4	-
80	78	1.0	90
90	490	6.1	160

Table 1: Increase of a parameter and angular size of depleted region with rising inclination.

tion δx can be calculated as [10]

$$\delta x = r_g \left[1 - \sqrt{1 - \frac{d^2}{r_g^2}} \right] \simeq \frac{d^2}{2r_g} \simeq \frac{eB_T c}{2E_\mu} d^2 \quad (10)$$

with r_g being the gyroradius of the muon.

Both relations can be combined into a factor a , which measures the amount of magnetic deformation.

$$a = \frac{\delta x}{r} = \frac{eB_T}{2p_T^\mu} d \quad (11)$$

If $a \geq 1$, two fully depleted regions will form along the direction of B_T (see Fig. 6) [10]. For $a < 1$, the former radially symmetric lateral muon distribution will be stretched perpendicular to B_T (compare with muon maps in Fig. 23 to Fig. 28 in the appendix).

Using $p_T \approx 300$ MeV, $B = 24.5$ μT and $d \approx d_{\max}$ from Fig. 4, we can calculate a for a 10^{19} eV shower at various zenith angles θ to get an impression how large the magnetic effects are. Because B_T will vary with azimuth, we will only use the maximum value at each inclination. In the case of $a \geq 1$, we can also define the angular size of the depleted regions α . Table 1 summarises the result.

Under realistic circumstances, the depleted regions are smeared out. This is related to the fact, that p_T and d form distributions and muon energies and directions are modified by ionisation loss and multiple scattering. Simulations therefore do not show sharply defined depleted regions, but in principle the consideration holds. The approximated values of α above $\theta = 80^\circ$ are roughly comparable with the simulation.

3.3 Early-late asymmetry

Independent of magnetic deformation, there is also a dipole-like asymmetry observed in simulations

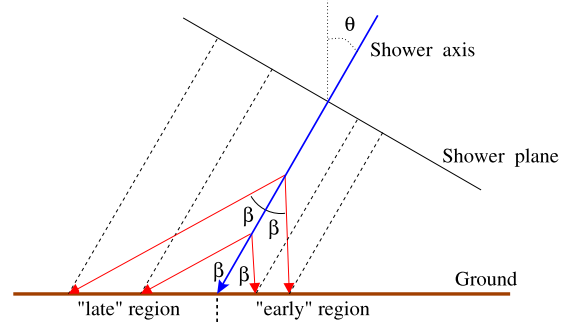


Figure 7: Schematic view of early and late parts (in terms of muon arrival time) of a shower that is hitting the ground. Muons produced under equal properties at a certain point in the shower development, arrive on the ground at different inclinations and after different path lengths (picture from [19]).

(compare Aachen muon map in Fig. 23 and Fig. 24 in the appendix) and data [19]. Tanks, which are hit early (in terms of shower arrival time) show higher signals than those in the late part. This is called the *Early-late* asymmetry. For moderate inclinations, this cannot be an effect of muon attenuation, and the reason is in fact more subtle.

The Early-late asymmetry appears, because muons with equal properties at their production point are measured under different conditions on the ground (see Fig. 7). Three mechanisms contribute to this, which all enhance the signal measured in the early tanks [19]:

- Early muons appear as more dense on the ground because of the shower geometry.
- Early muons are less inclined, thus the effective tank area $A_{\text{tank}}(\theta)$ is usually larger (maximum value at $\theta \approx 23^\circ$, see Eq. 16).
- Early muons travel shorter paths, resulting in less absorption.

The last point is only important for very inclined showers with $\theta \gtrsim 80^\circ$, where the shower front appears already nearly flat, but differences in the traveled path lengths become large. For lower inclinations, the curved front is the main source of the effect.

Our simulations show, that the total effect on $\bar{N}_\mu^{\text{tank}}$ in a radial distance $r = 1000$ m is between 40 % and 50 % for $\theta \in [60^\circ, 80^\circ]$, increasing with r .

3.4 Influence of shower parameters

This section shall summarise, how a muon map $\bar{N}_\mu^{\text{tank}}(x, y)$ depends on basic physical parameters. This will point out, why it is sufficient to compare tank signals in a given event against a muon distribution simulated at a fixed primary energy E and mass M .

We will distinguish between the influence on the *shape* of the muon map (meaning the normalised distribution) and on its *amplitude*, which shall be defined as the integral

$$N = \frac{\int_x \int_y \bar{N}_\mu^{\text{tank}}(x, y) dx dy}{\int_x \int_y dx dy} \quad (12)$$

Table 2 summarises, whether the impact of an event parameter is strong or weak.

We will discuss the energy influence first. Simplified Heitler-model calculations [1] and simulations agree (see Fig. 9), that N_μ^{ground} and the amplitude increase as a power of the energy

$$\text{amplitude} \propto N_\mu^{\text{ground}} \propto E^{1/\gamma} \quad (13)$$

with $\gamma \gtrsim 1$.

Now a main feature of the muon map $\bar{N}_\mu^{\text{tank}}(x, y)$ is the insensitivity of its shape to the primary energy E [20]. The large number of interactions that happen in the shower decouple the muon properties from the energy dependent first interactions to a high degree. The first interactions are responsible for the fluctuation in the total muon number but do not change the distributions.

The conditions at the shower maximum, where most muons are produced, are more important for the shape than the first interactions.

Cosmic rays with higher energies reach their shower maximum deeper in the atmosphere (see Eq. 7). The matter density ρ_{max} at this point has an impact on the muon energy distribution and therefore on the shape (see Section 3.1, p. 5). The distance to the ground along the shower axis d_{max} also modulates the energy distribution, the divergence and the geomagnetic deformation. But both ρ_{max} and d_{max} vary only slowly with energy E . In

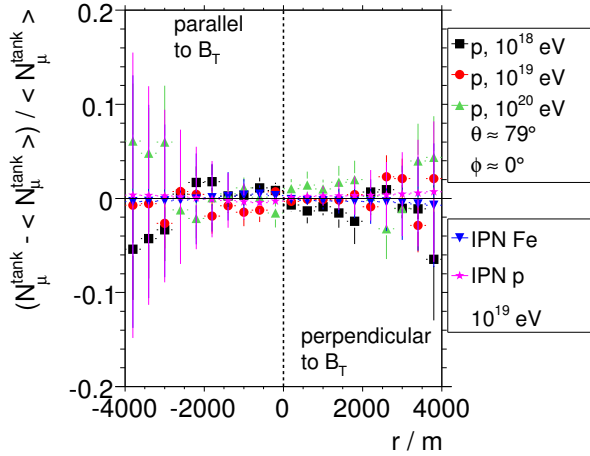


Figure 8: Deviation from an average muon count per tank along a line parallel and perpendicular to B_T in the shower plane for different primary energies and masses. To increase statistics, the first three graphs are each an average of ten normalised CORSIKA proton showers (see Table 4), taken from the interval $\theta \in [78^\circ, 80^\circ]$ and $\phi \in [-4.3^\circ, 5.7^\circ]$ and $\phi \in [175.7^\circ, 185.7^\circ]$. The bars show the statistical uncertainty of the deviation. The last two graphs compare independently a normalised proton and iron muon map generated by the Paris group [11], both simulated at 10^{19} eV. The uncertainty bars here are calculated using poisson statistics.

Parameter		Effect on	
		amplitude	shape
primary energy	E	+	-
primary mass	M	+	-
zenith angle	θ	+	+
azimuth angle	ϕ	-	+
geomagnetic field direction	$\mathbf{e}_B$	-	+
geomagnetic field strength	B	-	+

Table 2: Influence of physical parameters on shape and amplitude of a muon map $\bar{N}_\mu^{\text{tank}}(x, y)$.

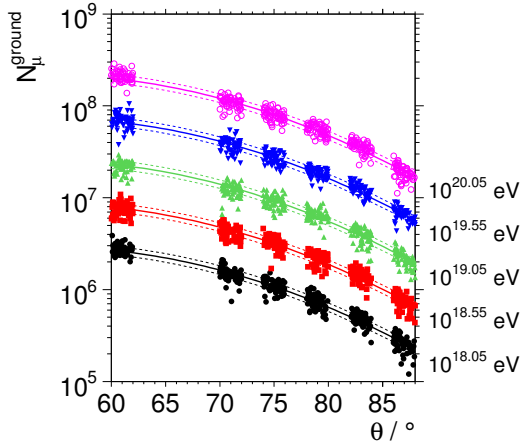


Figure 9: Total muon number on the ground as a function of zenith angle θ and primary energy E . The data points are taken from the simulations in Table 4. The energy E is not fixed in this plot. Instead, each value to the right of the plot represents the mean $\bar{E}$ of a narrow E^{-1} distribution with $\Delta \lg(E/\text{eV}) = 0.1$. The point spread is therefore in part due to shower-to-shower fluctuations (13 %, indep. of (E, θ) for $\theta \in [60^\circ, 88^\circ]$), in part due to the energy distribution. The continuous lines are fits of the form $\lg(N_\mu^{\text{ground}}) = c_0 + 1/\gamma \cdot \lg(E) + c_1 \cdot \theta + c_2 \cdot \theta^2$. The dashed lines indicate the expected spread due to the primary energy interval alone.

our simulations between $\theta = 60^\circ$ and $\theta = 80^\circ$, changes in $\bar{N}_\mu^{\text{tank}}(x, y)$ are only at the 5 % level. This small percentage is totally negligible compared to other effects as we will see later.

A similar argument goes for the primary mass M . Above 10^{18} eV, we can safely neglect the binding energy of the nucleus and approximate an iron shower as a *superposition* of 56 proton showers, each with a $1/56$ fraction of the total energy. Because the energy per nucleon is lower, iron showers have a higher shower maximum. On the other hand, they produce more muons compared to a proton of equal energy, because the muon number in a shower increases more slowly with energy than the photon or electron number [1]. Therefore, the mass M modulates the amplitude, but the effect on the shape is also negligible (see Fig. 8).

The insensitivity to mass M and energy E are exploited for the energy reconstruction algorithm: it is only necessary to simulate the muon map $\bar{N}_\mu^{\text{tank}}(x, y)$ for one particular combination. The amplitude of a real event can be measured by fitting the muon map to the data with a free normalisation factor, called N_{19} . If N_{19} is calibrated against a composition-unbiased energy scale (like the FD scale), the reconstructed energy will also be independent of composition assumptions.

Amplitude and shape of muon maps are very sensitive to the inclination θ of the shower, with the sensitivity growing for larger zenith angles. This can be understood by looking again at Fig. 4: correlated to the distance between shower maximum and surface d_{max} and the integrated atmospheric depth X_{atm} are the amount of geomagnetic deformation a (see Eq. 11), divergence (see Eq. 9) and absorption (see Fig. 9).

The azimuth angle ϕ modulates the geomagnetic

deformation a , because B_T is dependent on the orientation between shower axis and geomagnetic field (see Eq. 11). The amplitude however remains unaffected.

Naturally, the direction and magnitude of the geomagnetic field itself also change the shape through the magnitude of B_T (see Eq. 11). However, since the magnetic field is expected to vary only slowly as stated in Section 3.2, p. 7, it can be considered constant.

More details on these topics can be found in [3, 10, 21].

3.5 Available muon maps

The last sections showed, that the mean muon number per tank $\bar{N}_\mu^{\text{tank}}$ is a function of the shower plane coordinates (x, y) (see Fig. 2) and all primary shower parameters. It also showed, that the dependency on E and M can be factorised.

$$\begin{aligned}\bar{N}_\mu^{\text{tank}} &= \bar{N}_\mu^{\text{tank}}(x, y; \theta, \phi, E, M) \\ &\simeq N_{19}(E, M) \\ &\quad \times \tilde{N}_\mu^{\text{tank}}(x, y; \theta, \phi, 10^{19} \text{ eV}, M_p)\end{aligned}\quad (14)$$

The muon map is simply the last part of this equation. In the following, we will omit the implicit energy and mass dependency and just refer to the muon map by:

$$\bar{N}_\mu^{\text{tank}} = \bar{N}_\mu^{\text{tank}}(x, y; \theta, \phi) \quad (15)$$

$\bar{N}_\mu^{\text{tank}}$ can be converted into a tank signal in a straight forward way, because the signal in a Water-Cherenkov-Detector is mainly only a function of the path length of the muon in water and independent of the muon energy. In contrast, the signal of electromagnetic particles is very energy dependent, because of the production of sub-showers and secondary light inside the water volume.

The energy reconstruction algorithm usually uses the muon number per tank to calculate and look up the expected muon signal distribution for each tank individually. The signal distribution is a poisson distribution convoluted with the expected (simulated) signal distribution for a single muon. For the proper calculation of the signal probability distribution, it is thus more reasonable to parameterise the muon number per tank and not the muon signal directly.

We now want to try to give an overview of the available muon maps, which were made by different groups and by using different shower Monte-Carlos and refinement methods. Table 3 summarises their individual features.

The two groups of the University of Santiago de Compostela [10] and the IPN Paris [11] used the shower Monte-Carlo code AIRES [12] together with the hadronic interaction model QGSJet01 [13] for the muon map generation. They differ in the way, how muons from the simulation are treated after their production and how the muon densities are refined into a final parameterisation. Both map types have certain advantages and disadvantages. For example, there are no Early-late asymmetries included into the Santiago map, while the Paris map was only generated for $\theta \geq 70^\circ$.

The third and new muon map from our group tries to combine the advantages on the one hand, and be complementary in the choice of the shower simulation the other hand. Instead of AIRES we used the CORSIKA Monte-Carlo [4], which is an important cross check. We also used another interaction model: QGSJet-II [5] is the successor of QGSJet01 and introduces non-linear effects to pomeron interactions, among other improvements. It is expected to give a more realistic description of the muon component. A comparison done in [3] however showed, that the lateral muon number distribution apart from a global factor appears to be nearly unchanged by the choice of QGSJet-II over QGSJet01. The full availability of the electromagnetic distribution on the ground is another advantage in our shower library.

3.6 Aachen muon map

To generate a continuous and average description of the mean muon number per tank $\bar{N}_\mu^{\text{tank}}(x, y, \theta, \phi)$ out of a library of Monte-Carlo showers, some refinement steps are necessary.

The particle files from individual showers of the simulation need to be binned and the muon number density converted into a muon count per tank, using the effective tank area of an Auger tank. This already provides a raw description of $N_\mu^{\text{tank}}(x, y)$ for every single shower, which still includes all individual poisson artificial fluctuations. Shower-to-shower fluctuations are also present, but they generally affect only the total number of muons in a

Group	USC Santiago	IPN Paris	RWTH Aachen
Shower Monte-Carlo	AIRES	AIRES	CORSIKA
High energy interaction model	QGSJet01	QGSJet01	QGSJet-II
Primary particles	protons	protons and iron	protons
Primary energy	10^{19} eV	10^{19} eV	10^{19} eV (scaled ^a)
Zenith angle range	$60^\circ - 89^\circ$	$70^\circ - 89^\circ$	$60^\circ - 88^\circ$
Early-late asymmetry	no	yes	yes
Steps in θ, ϕ	$2^\circ, 5^\circ$	continuous	continuous
Steps in x,y	20 m, 20 m	continuous	continuous
Unit	muons / m ²	muons per tank	muons per tank
Showers used per map	100	50	-/global fit ^b
Muon map generation method	Simulation without magnetic field, deformations added by semi-analytical model. [10]	Separated Monte-Carlo for muon propagation. Parameterisation of ground densities. [11]	Full shower simulation within CORSIKA. Parameterisation of ground densities.

^aShowers in an energy range of 10^{18} eV to $10^{20.1}$ eV were used. Details in Section 3.6, p. 11.

^bCannot be compared directly. To average shower-to-shower fluctuations, a global fit to all showers at once was used. See Section 3.6, p. 11.

Table 3: Available muon maps, generated by different groups

shower. Using $\bar{N}_\mu^{\text{tank}} \propto E^{1/\gamma}$, it is also simple to scale the muon count per tank to the equivalent at a primary energy of 10^{19} eV, if necessary.

The fluctuations and the discreteness of the raw showers result in the need of a method to average and interpolate available data.

We chose to follow ideas presented in [11] to address both challenges. Essentially, we fit a parameterisation to each individual $N_\mu^{\text{tank}}(x, y)$ distribution, which results in a set of parameters a_{kj}, b_{kj} for each individual shower. Then we fit a second parameterisation to these shower parameters, which tries to describe their (θ, ϕ) evolution. While the first parameterisation flattens poisson and artificial fluctuations^b) on the shower level, the second parameterisation averages the shower-to-shower fluctuations and interpolates between known values of (θ, ϕ) .

Our input set is a general purpose library of 1800 CORSIKA proton showers [3]. The distribution of the showers inside the library is summarised in Table 4.

^b)Artificial fluctuations are introduced by the thinning method in the simulation.

3.6.1 Parameterisation and Fitting

This subsection will explain the parameterisation and fitting procedure in technical detail. It is strongly based on [11], but has some key differences. The procedure follows four steps. The first three steps are applied on each individual shower, the last one is applied to the set as a whole.

1. **$N_\mu^{\text{tank}}(\sqrt{r}, \psi; \theta_i, \phi_i)$ calculation.** Muon coordinates from the ground plane are projected into the shower plane and converted into the polar coordinates $\sqrt{r}, \psi$. Choosing $\sqrt{r}$ over r will improve the fit later, because of the experimental observation that approximately $\lg \bar{N}_\mu^{\text{tank}} \propto \sqrt{r}$. The $\sqrt{r}, \psi$ -plane is divided into a grid, and the muon number density $n_\mu(\sqrt{r}, \psi)$ calculated in each cell. The same projection and binning is also applied to the local muon inclination $\cos \theta_\mu$, which varies up to $\Delta \cos \theta_\mu = 0.5$ in a single shower at $\theta = 60^\circ$.

The muon number density $n_\mu(\sqrt{r}, \psi)$ is converted into a muon count per tank $N_\mu^{\text{tank}}(\sqrt{r}, \psi)$ by using the effective area of an

Parameter	distribution	bin width	low edge of bin
$\lg E/\text{eV}$	flat	0.1	18.0, 18.5, 19.0, 19.5, 20.0
$\theta/^\circ$	$\sin(\theta)\cos(\theta)$	2	60, 70, 74, 78, 82, 86
$(\phi + \delta)/^\circ$	flat	10	0, 30, 60, 90, 120, 150, 180, 210, 240, 270, 300, 330

Table 4: Parameter distribution of the CORSIKA/QGSJet-II/FLUKA proton shower library [3–6]. $\delta \approx -4.3^\circ$ is the current magnetic declination in Malargüe. The library contains five showers per bin and 1800 showers in total.

Auger surface tank in the shower plane:

$$\begin{aligned}
A_{\text{tank}}(\theta_\mu) &= A_{\text{top}}(\theta_\mu) + A_{\text{side}}(\theta_\mu) \\
&= \pi r^2 \cos \theta_\mu + 2rh \sin \theta_\mu \quad (16) \\
&\text{with } r = 1.2 \text{ m} \quad h = 1.8 \text{ m}
\end{aligned}$$

As $\cos \theta_\mu$ varies within the shower, the effective area also varies. Muons in the early part of the shower are less inclined and see a larger effective tank area. This is partly responsible for the Early-late asymmetry.

2. **Energy Scaling.** $N_\mu^{\text{tank}}(\sqrt{r}, \psi)$ is not yet scaled to an equivalent energy of $E = 10^{19}$ eV. We exploit the scaling law of the total muon number on the ground to the primary energy: $N_\mu^{\text{ground}} \propto E^{1/\gamma}$. γ is fitted from the simulation. For the proton CORSIKA/QGSJet-II/FLUKA library, we obtain the scaling relation:

$$\begin{aligned}
\frac{N_\mu^{\text{tank}}(E)}{N_\mu^{\text{tank}}(10^{19} \text{ eV})} &= \left(\frac{E}{10^{19} \text{ eV}} \right)^{1/\gamma} \quad (17) \\
\gamma &= 1.069 \pm 0.038 \text{ (statist.)}
\end{aligned}$$

3. **Parameterisation on shower level.** The raw distributions $N_\mu^{\text{tank}}(\sqrt{r}, \psi; \theta_i, \phi_i)$ still include statistical and artificial fluctuations from the simulation. A fit to a parameterisation is applied, which smoothes these fluctuations (see Fig. 21 in the appendix):

$$\begin{aligned}
\lg \bar{N}_\mu^{\text{tank}} &= \sum_{k=0}^3 \sqrt{r}^k \times \\
&\left(\sum_{j=0}^3 a_{kj} \cos(j\psi) + \sum_{j=1}^3 b_{kj} \sin(j\psi) \right), \quad (18)
\end{aligned}$$

leading to 28 parameters for each map. For moderate magnetic deformations between

$60^\circ < \theta < 80^\circ$, the quality of the fit is very good. Above 80° , fluctuations due to low particle statistics make the fit more uncertain, especially in the depleted regions (see Fig. 6).

Important note: Only bins with $150 \text{ m} < r < 4000 \text{ m}$ are included in the fit. Near to the core, the simulation introduces artificial fluctuations, which are too large for a reasonable fit, caused by a radial thinning in the shower simulation (the top left plot in Fig. 21 in the appendix documents the effect). Bins above $r = 4000 \text{ m}$ are excluded, because the particle density becomes too low.

The 28 parameters are saved together with the values of zenith and azimuth angle for each shower.

4. **Parameterisation on global level.** The parameters of the last step are dependent of zenith θ and azimuth angle ϕ . A second parameterisation attempts to approximate the evolution of these parameters with the incoming direction of the shower:

$$\begin{aligned}
a_{kj}, b_{kj} &= \sum_{m=0}^3 \tilde{d}_{\text{max}}(\theta)^m \times \\
&\left(\sum_{l=0}^2 c_{kjml} \cos(l\phi) + \sum_{l=1}^2 d_{kjml} \sin(l\phi) \right) \quad (19)
\end{aligned}$$

resulting in a final set of $28 \times 20 = 560$ parameters (some fit examples are shown in Fig. 22 in the appendix).

$d_{\text{max}}(\theta)$ is the distance between the shower arrival point on the ground and the shower maximum (see Fig. 3). It is roughly the scale on which the magnetic deflection, divergence and attenuation grows. $\tilde{d}_{\text{max}}(\theta)$ is an approximation to this value, calculated from Eq. 6 by assuming a constant average height of the shower

maximum $h_{\max} = 15$ km. $h_{\max}$ itself is a slowly varying function of θ and energy (see Fig. 4), but for the fit we simply set it constant. Using $\bar{d}_{\max}(\theta)$ over a simple power of θ improves the fit considerably.

The set of 560 parameters of first and the second fit together provide an averaged description of the lateral muon distribution at 10^{19} eV for all interesting incoming directions, which completes our task.

Compared to [11], we did not imply a symmetry $N_{\mu}(x, y; \theta, \alpha) = N_{\mu}(x, y; \theta, -\alpha)$ on the second fit, with α being the azimuth angle relative to the ground projection of the magnetic field. For $\alpha \rightarrow -\alpha$, muons and anti-muons change place in the muon map. If muons and anti-muons would behave the same way in the atmosphere, this symmetry would be exact. In reality, the absorption for muons is slightly larger. This results in different shapes, depending on whether muons or anti-muons are in the early or late part of the shower. The effect is small, but we found that our fits improve without imposing the symmetry (see Fig. 22 in the appendix).

4 Comparisons

In this section, we compare the different muon maps, summarised in Table 3. A muon map can have two effects on the reconstructed energy (see Fig. 10). On the one hand, it can introduce an offset of the reconstructed energy against the true energy. We will see later, that this is well compensated through the FD hybrid calibration. On the other hand, the muon map reconstruction introduces a limited energy resolution. The best energy resolution can be achieved with a muon map that fits perfectly to the data.

Because we don't know the true energy in our analysis, we can only do relative comparisons between the three maps. But by looking at a shift of the average energy from choosing one muon map over another, we will estimate the size of a possible systematic offset. We will also take a look at the energy fluctuation caused by choosing one map or another. Further, we try to find the map that best matches the signal profile in real events.

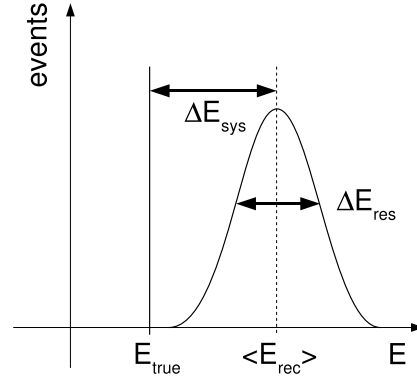


Figure 10: Possible effects of a muon map on the reconstructed energy. ΔE_{sys} denotes a systematic shift between the average reconstruction energy and the true energy. ΔE_{res} is the limited energy resolution introduced by the map reconstruction.

4.1 Map-to-map comparison

We study all three map approaches in their overall amplitude and shape, which both affect the reconstructed energy.

4.1.1 Amplitude

For the energy reconstruction with N_{19} , a constant factor between the amplitude of two different map types is not relevant for the energy reconstruction. It would be simply absorbed in the calibration of N_{19} against the Fluorescence Detector energy E_{FD} . If instead the amplitude has a different evolution with θ , it will systematically affect the energy reconstruction. We remind here, that the energy is calibrated mostly by events with zenith angles lower than 70° [17]. For higher zenith angles, this calibration is effectively extrapolated using the muon map evolution of the amplitude with zenith angle. This may introduce a bias, if the attenuation with θ is not correctly modelled into the map.

To compare the amplitude differences, all maps are first converted into the same format. The number of muons per tank is calculated and binned in a square of ± 4000 m in (x, y) with a step size of 20 m. This choice is motivated by a wish for compatibility with the already binned Santiago format.

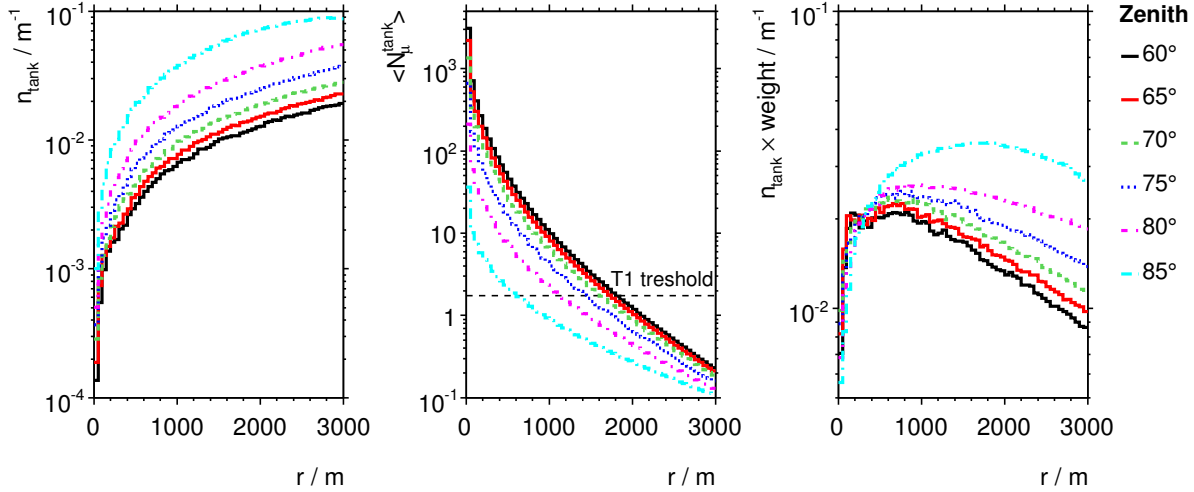


Figure 11: From left to right: a) Monte-Carlo calculated radial tank density for a perfect Auger grid, b) muon number per tank using the Aachen muon map, c) significance of radial distance for a fit. The significance in each bin is defined as tank density weighted by the inverse relative poisson uncertainty $\left(\sqrt{N_{\mu}^{\text{tank}}/N_{\mu}^{\text{tank}}}\right)^{-1} = \sqrt{N_{\mu}^{\text{tank}}}$. r is the distance from the shower axis.

Then, an attempt is made to compare the amplitude only for that region of a muon map, which is most important for the energy reconstruction. Because of the rapid decrease of the muon density with increasing distance to the core, a simple integral over each map is dominated by values near the core. In a fit against real data on the other hand, the map amplitude is mostly constrained by signals between $200 \text{ m} \lesssim r \lesssim 2000 \text{ m}$ (see Fig. 11).

Therefore, we compare the integrated number of N_{μ}^{tank} in these radial bounds^{c)}.

$$N_i = \frac{\int_{200 \text{ m}}^{2000 \text{ m}} dr \int_0^{2\pi} d\phi N_{\mu}^{\text{tank}}(r, \psi; \theta, \phi)}{\int_{200 \text{ m}}^{2000 \text{ m}} dr \int_0^{2\pi} d\phi} \quad (20)$$

Fig. 12 shows the relative difference in amplitude between all pairs as a function of zenith and azimuth angle. The dominant effect is the variation of the difference with zenith angle. The Aachen map expects the value of N_{19} to decrease faster above $\theta \gtrsim 72^\circ$ than Paris or Santiago. The opposite is true for angles below 72° compared with Santiago.

^{c)}The result was checked to be insensitive against small variations of the radial bounds. For an inner bound below $r = 50 \text{ m}$, this is no longer true.

In the case of Santiago against Paris, there is only a weak trend. The variation of the relative difference between the two is smaller, 10 % compared to 16 % (Santiago/Aachen) and 13 % (Paris/Aachen).

The variation with azimuth in all pairs is even smaller, in the order of 5 %.

We conclude, that the Aachen map shows a different evolution of N_{19} with the zenith angle than Paris and Santiago. The effect is of the order of 20 %, resp. 24 %.

The effects discussed here will show up again in Section 4.3.3, p. 23 and Section 4.3.4, p. 23.

4.1.2 Shape

The Monte-Carlo simulations and the approaches used to generate the different map types lead to slightly different shapes. These differences will affect the fit quality at best, but they can in principle also lead to a systematic offset in the reconstructed energy or a decreased energy resolution. The reconstructed energy of a wrongly shaped muon map will depend on the particular tank topology in a each event.

We start our comparison again from the common map format discussed in Section 4.1.1, p. 14.

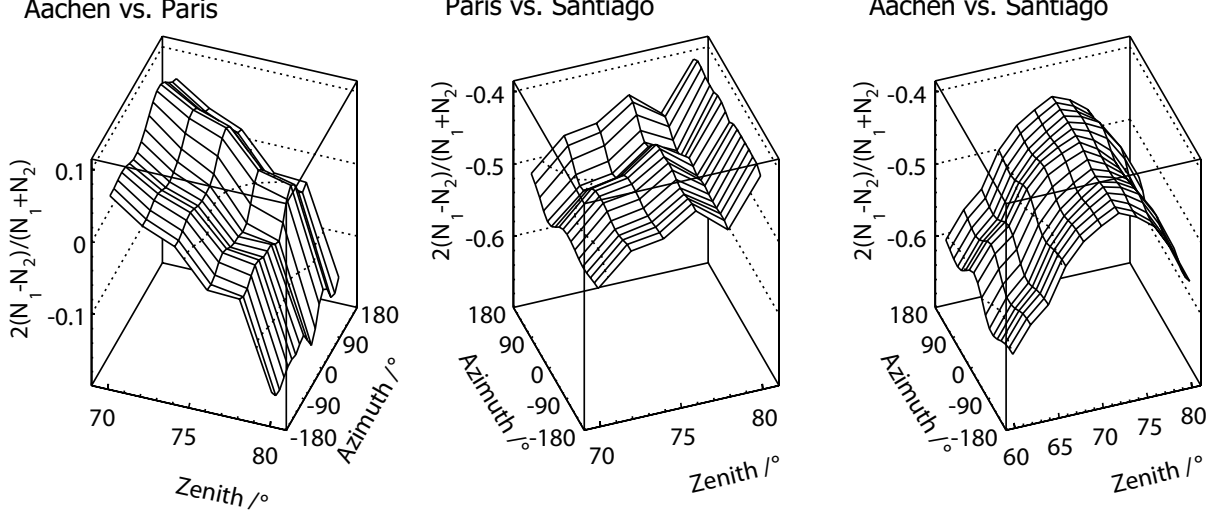


Figure 12: Relative difference in the amplitude N for a pair of maps. N is the integrated muon number per tank in the radial distance $200 \text{ m} < r < 2000 \text{ m}$ from the shower axis. An explanation for this special choice of N is given in the text.

To show only differences in shape, we normalise all maps to the Santiago map using the amplitude N defined in Eq. 20:

$$\tilde{N}_{\mu,i}^{\text{tank}} = \frac{N_{\text{Santiago}}}{N_i} \tilde{N}_{\mu,i}^{\text{tank}} \quad (21)$$

Fig. 23 to Fig. 28 in the appendix show these comparison plots for all pairs at the most significant zenith and azimuth angles. The relative difference in the predicted number of muons per tank is shown as well as the difference divided by the expected local Poisson fluctuation.

$$X_1 = \frac{N_{\mu,i}^{\text{tank}} - N_{\mu,j}^{\text{tank}}}{\frac{1}{2}(N_{\mu,i}^{\text{tank}} + N_{\mu,j}^{\text{tank}})} \quad (22)$$

$$X_2 = \frac{N_{\mu,i}^{\text{tank}} - N_{\mu,j}^{\text{tank}}}{\sqrt{\frac{1}{2}(N_{\mu,i}^{\text{tank}} + N_{\mu,j}^{\text{tank}})}} \quad (23)$$

Since fits essentially try to minimise something like

$$\frac{\text{data} - \text{model}}{\text{uncertainty}}$$

X_2 provides a better measure on how important the differences are for the energy reconstruction. The downside of X_2 is, that it is energy dependent. We regard X_2 for an energy of 10^{19} eV , at 10^{20} eV (10^{18} eV) X_2 is larger (smaller) by a factor of 3.

The relative difference X_1 and the difference divided by the Poisson uncertainty X_2 give a very different picture.

The relative difference X_1 in parts of the muon map can easily grow over 100 % for most shower directions. In all cases, Aachen expects a steeper fall-off of the muon number per tank here than the other maps. In general, Paris and Santiago agree better in a comparison based on X_1 , except for cases with strong magnetic deformations ($\theta > 74^\circ$, ϕ around 0° or 180°).

The difference divided by the local Poisson uncertainty X_2 does not exceed $\pm 3\sigma$, except very close to the shower axis. In general, Aachen and Paris agree better in a comparison based on X_2 . In cases with strong magnetic deformations, both maps expect a sharper and steeper drop in the muon density along the direction of $\mathbf{B}_T$ than the Santiago map. Regions with large deviations grow with the amount of magnetic deformation.

The largest differences in X_2 appear close to the shower axis, where the Poisson uncertainty is also small. Paris and Aachen both expect fewer particles near the axis than Santiago, with deviations above 5σ . However, deviations below $r = 200 \text{ m}$ are expected to affect the reconstructed energy only weakly (after the considerations in Section 20,

p.15).

The difference in including the Early-late asymmetry into the muon map or not is best seen at $\phi = 0^\circ$. The Santiago map is right-left symmetric, while the Paris and Aachen maps are not. The Early-late asymmetry causes differences in muon numbers up to 2σ , with the largest influence at low inclinations near $\theta = 60^\circ$.

We conclude, that the shapes of all candidates differ significantly, with systematic shifts in N_μ^{tank} of over 100 % or $\pm 3\sigma$ of a Poisson distribution at 10^{19}eV . Including the Early-late asymmetry into the muon map gives a significantly different shape and should not be neglected.

4.2 Map shape compared with real data

4.2.1 Data set and reconstruction software

For the following analyses, we use Surface Detector data from 2004/01/01 to 2007/03/01 with a total cleaned exposure^{d)} of $2287\text{ km}^2 \cdot \text{sr} \cdot \text{y}$ for $\theta \in [60^\circ, 80^\circ]$ [22].

Physical events are selected by the official T4HAS [23]. It is a seedless trigger approach that basically takes the maximum number of tanks into account, that show a reasonable time-position correlation with a flat shower front. High quality events are selected with the standard T5 cut, which requires the tank with the highest signal in the event to be surrounded by a full hexagon of active stations.

The selected events are reconstructed using a slightly modified version of the `efit-v4r2p4` software of the University of Santiago de Compostela [17]. In order to use the Aachen and Paris maps with this software, we convert all maps into the common format previously described in Section 4.1.1, p.14. Additionally, the Paris and Aachen maps are discretised in the θ, ϕ -plane in the same way like the Santiago map (see Table 3). In the case of the Paris map, only maps above $\theta = 70^\circ$ are available.

To use the common map format, two minor modifications to the `efit` code are necessary.

^{d)}Based on the T5 criteria: To count an elementary grid cell at a given time, the central tank and its six surrounding stations need to be active and taking data. Time periods with an uncertain central trigger accuracy are removed from exposure and data.

- All internal conversions of the type $N_\mu/\text{m}^2 \rightarrow N_\mu/\text{tank}$ are removed.
- The routine to select a map according to a (θ, ϕ) pair is modified to also handle maps that include asymmetries.

Using the selections from above, we reconstruct three data sets. In the zenith angle range between $\theta = 60^\circ$ (70° for Paris) and $\theta = 80^\circ$ we get

- 32877 events using the Santiago map,
- 13436 events using the Paris map,
- and 33497 events using the Aachen map.

The lower number of events for the Paris map reflects the smaller exposure fraction of its defined θ -range. The event numbers of the Santiago and Aachen map differ by 2 %. An analysis shows, that the surplus in the Aachen map data set consists entirely of low energy events. This might indicate, that the muon map fit converges better at low energies with the Aachen map. However, since these events are all below the SD trigger saturation threshold and therefore irrelevant to the energy spectrum, we did not investigate this further.

To see how well the different muon maps describe the measured tank signal profile in real events, we examined the fitting quality of each muon map using the χ^2 probability distribution and the distribution of average residuals.

4.2.2 χ^2 probability analysis

To compare our modelled muon maps with real data, we can use the χ^2 test. The χ^2 value is calculated as follows:

$$\chi^2 = \sum_i^N \left(\frac{S_i^{\text{measured}} - S_i^{\text{model}}}{\sigma_{S,i}} \right)^2 \quad (24)$$

where the fraction in the sum is the difference between measured and modelled signal divided by the expected signal uncertainty. χ^2 is calculated on an event-per-event basis after the muon map fit (with fixed θ and ϕ).

If $S^{\text{measured}} - S^{\text{model}}$ follows a gaussian with zero mean and variance σ_S , the χ^2 will follow the known χ^2 distribution after the reconstruction of many events:

$$f(x; k) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} \exp^{-x/2} \quad (25)$$

where $x = \chi^2$ and k is the number of degrees of freedom. After the muon map fit to the data, we get the total normalisation of the map N_{19} and two ground coordinates of the core as derived parameters, so $k = N_{\text{stations}} - 3$.

We can now calculate the per-event probability p , to find a χ^2 equal or larger than the actual one.

$$p = \int_{\chi^2}^{\infty} f(x; N_{\text{stations}} - 3) dx \quad (26)$$

If all assumptions for the χ^2 -statistic are fulfilled, p will follow a uniform distribution between $[0, 1]$.

We use the `efit` software to calculate the χ^2 for each event and then calculate p accordingly. Fig. 13 shows the p distributions for a set of "benchmark points" in the θ, ϕ -plane. These points are actually small regions around a certain point to gather statistics^{e)}.

We observe, that for all benchmark points the real distribution is not uniform and the mean is between 0.15 and 0.25, which is too low. We can now try to understand this. Eq. 24 shows, that there are three factors which influence the χ^2 distribution:

- the modelled signal S_i^{model} (correct mean value),
- the expected uncertainty $\sigma_{S,i}$ (correctly expected fluctuation around mean value),
- and the distribution of $S_i^{\text{measured}} - S_i^{\text{model}}$ (gaussian or non-gaussian).

While we don't know, whether our model of the mean signal is correct, we can with certainty say that the expected signal uncertainty is not. As we pointed out already in Section 3.1, p. 5, the expected signal uncertainty in the `efit` software is too low, because fluctuations of the residual electromagnetic signal component are not taken into account.

There is another point about fluctuations. Our data is totally dominated by low signals of only a few muons per tank. The χ^2 -test expects the real fluctuation to be of gaussian shape. In case of Poisson statistics, this is only a good approximation for $N_{\mu}^{\text{tank}} \gtrsim 10$. At low signals, the fluctuations of S_i^{measured} turn out to artificially decrease the χ^2 , which makes p larger.

^{e)}Their centers and sizes are shown in the plot legend.

A toy Monte-Carlo showed, that the observed χ^2 distribution can be reproduced with a correct muon map and the two described fluctuation related effects alone.

We therefore conclude at this point, that the χ^2 distribution is not a good statistical tool to give an absolute quality measurement of the muon map. Nevertheless, we can attempt a relative comparison of the fitting quality, because the fluctuation effects are essentially the same for all maps.

While the difference between the individual maps is small against the total deviation from 0.5, the test slightly favors the Paris and Aachen maps over the Santiago map. It is reasonable to assume, that the simplifications in the Santiago map like the missing Early-late asymmetry are causing this.

4.2.3 Residual analysis

Because the χ^2 test cannot decide well between a wrong model and wrongly estimated signal uncertainties, we also take a look at the residuals R_i of measured signal and muon map model

$$R_i = \frac{S_i^{\text{measured}} - S_i^{\text{model}}}{\sigma_{S,i}} \quad (27)$$

where $\sigma_{S,i}$ is the expected signal uncertainty. Because we use the `efit` software to calculate $\sigma_{S,i}$, only the Poisson uncertainty of N_{μ}^{tank} is considered. The fluctuation of the electromagnetic signal contribution is neglected (see Section 3.1, p. 5).

Using our large set of events, we can bin the average residual in the r, ψ -plane. This will show systematic deviations from the muon map, which are more independent of a wrongly estimated signal uncertainty. Care has to be taken for saturated tanks, which we exclude from this analysis by making a very conservative cut on the measured signal: $S_i^{\text{measured}} < 100 \text{ VEM}$.

Fig. 29 to Fig. 31 in the appendix show the result. The most striking feature is the overestimation of the real signal near the shower axis at all angles in all maps up to 9σ , especially above $\theta = 70^\circ$. The (necessary) opposite effect is found around $r = 1000 \text{ m}$, where we find a consistent underestimation of the measured signal in all maps and all directions up to 2σ .

Both effects are more pronounced in the Santiago map than in the Aachen and Paris maps, which give nearly the same residuals. The best agreement with

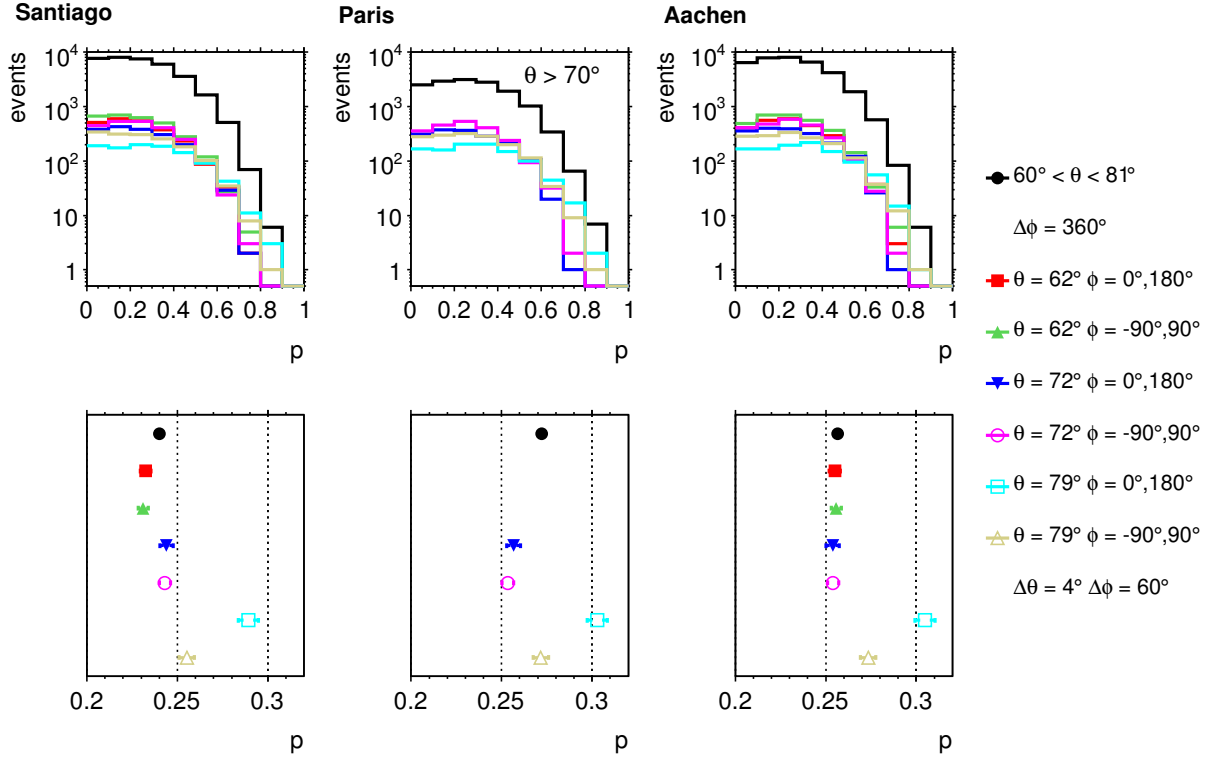


Figure 13: Quality of the muon map fit in a T5 selected set of inclined data. The top row shows the distribution of the χ^2 probability p for all selected events (see text). It is the per-event-probability to get a χ^2 larger than the one of the fit just by chance fluctuations. The ideal distribution of p would be flat in the interval $[0, 1]$ with an average of 0.5. The bottom row of plots shows the mean and its uncertainty from the distributions above. The colors denote different regions in the θ, ϕ -plane. The legend shows the mean and interval of each region. In case of the Paris map, no data below $\theta = 70^\circ$ is available.

data is found at $\theta = 79^\circ$, but we have to keep in mind, that the signals in this region are lower and the estimated uncertainties therefore larger, which already results in smaller residuals.

The residuals were also binned in ψ , the angle in the shower plane (see Fig. 2). If the magnetic distortion is correctly simulated in the muon map, the residual distribution $R(r, \psi)$ has to be independent of ψ . In contrast, we can see a dependency on ψ for $\theta > 70^\circ$ and $\phi = 0^\circ$ (180°), where we get strong magnetic field effects. This ψ -modulation is again larger for the Santiago map, up to 1.5σ , and smaller/comparable for the Aachen and Paris maps. In particular, the Santiago map underestimates the magnetic distortion, while the latter two overestimate it slightly.

At $\theta = 62^\circ$, the Aachen map gives a better description of the real signal profile than the Santiago map, which shows the improvement of including the Early-late asymmetry into the muon map.

4.3 Map influence on the reconstructed energy

4.3.1 Data set and reconstruction software

For the Surface Detector, we start from the same reconstructed data sets as in Section 4.2.1, p. 17. For the Fluorescence Detector hybrid data, we use the ADST event tree of the Auger Observer group [24], which is reconstructed using the Offline-v2.3 software [7].

4.3.2 Energy calibration

We want to compare the energy distributions resulting from different muon maps, instead of the rather biased value of N_{19} . To derive the reconstructed energies, we calibrated the data set for each muon map individually against a selection of high quality FD hybrid events.

To select only high quality events for the calibration, we started with common cuts from [17] and adapt them for the use with inclined showers:

- use only events from Coihueco and Los Leones
- use only events which have a measurement of the Mie-scattering length
- shower maximum in the field of view

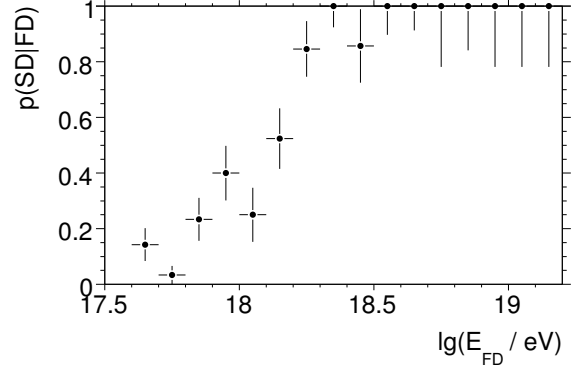


Figure 14: Trigger probability for the SD as a function of the FD energy. The measurement is done by using FD hybrids (see text). The probability saturates to one at an energy of $\lg E/\text{eV} \approx 18.5$.

- number of pixels in the track > 5
- track length $> 250 \text{ g cm}^{-2}$
- fraction of Cherenkov light up to 80 %
- χ^2/ndof of the Gaisser-Hillas fit < 2.5
- $\Delta E_{\text{FD}}/E_{\text{FD}} < 0.5$
- $\Delta X_{\text{max}} < 40 \text{ g/cm}^{-2}$
- χ^2/ndof of the muon map fit < 2

If a shower is recorded by two eyes and both FD reconstructions pass the quality cuts, the weighted average of the energies is used.

Fig. 15 shows the intersection of the FD hybrid data set and SD data set for each map. The uncertainties are those reported by the according reconstructions, with an additional uncertainty of 13 % added to N_{19} . This additional uncertainty accounts for shower-to-shower fluctuations of the total muon number and was derived from the CORSIKA proton shower library.

For the calibration, a potential law relation between FD energy E_{FD} and N_{19} , which is proportional to the total muon number on the ground, is fitted to the data:

$$E = E_{\text{MC}} \cdot N_{19}^\gamma \quad (28)$$

In this formula, E_{MC} can be interpreted as that measured FD energy, that produces the same amount of muons like the Monte-Carlo at 10^{19} eV .

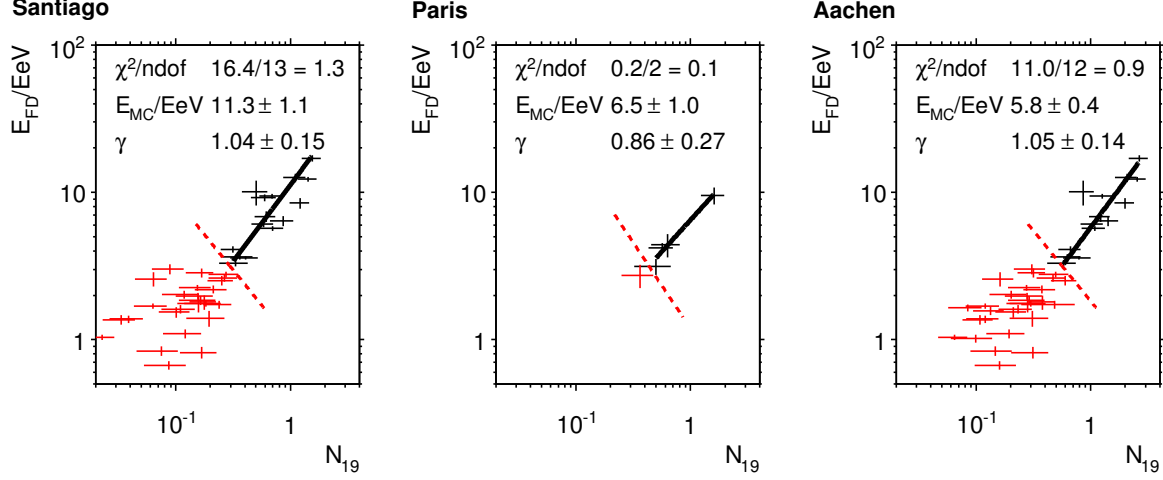


Figure 15: Energy calibration for N_{19} using hybrid events. The red points are below the SD trigger saturation threshold and excluded from the fit. The dashed red line shows the adopted energy cut (see text).

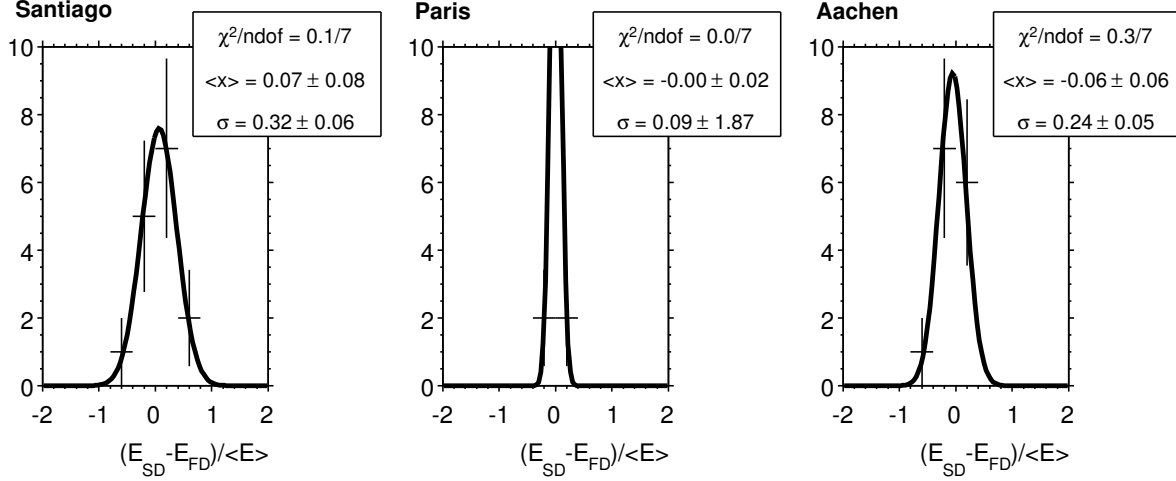


Figure 16: Energy resolution, derived from the calibration. For every point (N_{19}, E_{FD}) in Fig. 15, we calculate the SD energy from $E_{\text{SD}} = E_{\text{MC}} \cdot N_{19}^\gamma$ and then the relative difference to the FD energy $(E_{\text{SD}} - E_{\text{FD}})/E_{\text{FD}}$.

To avoid a bias, low energy inclined events that do not saturate the SD triggers have to be excluded from the fit. Below trigger saturation, the SD data set will contain more events that are upward fluctuations of the average N_{19} , thus introducing the bias. The cut has to be implemented as a line roughly perpendicular to the calibration curve, otherwise the cut itself would introduce a up- or downward pull on the calibration curve. The exact angle is dependent on the experimental uncertainties on N_{19} and E_{FD} and can in principle be calculated. However, our currently limited data set is not sensitive to moderate tilts of the cut line, therefore we simply use a perpendicular line at this point. The combination of cut and fit is found by an iterative procedure.

The position of the line at an energy of $\lg(E_{thres}/\text{eV}) = 18.5$ is deduced from Fig. 14. The plot shows the measured probability $p(SD|FD)$ for an FD hybrid shower to trigger the Surface Detector, as a function of the hybrid energy. The basic idea behind this is, that the conditioned probability $p(SD|FD)$ saturates to one, if the SD trigger chain saturates. To correctly calculate the probability, a cut on the FD hybrid data has to make sure, that the shower actually hits an active part of the SD array. We used a cut, that requires the station next to the reconstructed FD core to be surrounded by a hexagon of at least five working stations. The plot shows a well-defined saturation above $\lg(E_{thres}/\text{eV}) = 18.5$.

It is interesting to point out the contradiction between Fig. 14 and Fig. 1. The hybrid data implies a much lower saturation energy than the detector simulation using proton showers generated with QGSJet-II. This is somewhat expected, because all current shower simulations produce a lower amount of muons than what is observed in real data at the same primary energy [8]. Quantitatively, we can ask whether both numbers are compatible after correcting for the mismatch in the energy scales.

To correctly compare, we need to transfer the (wrong) Monte-Carlo energy scale to the FD energy scale. We exploit the fact, that the E_{MC} parameter from the calibration fit is that FD energy, which shows the same amount of muons on the ground than the Monte-Carlo at 10^{19} eV. This Monte-Carlo energy is by coincidence also the trig-

ger threshold found in the simulation.

$$\lg(E_{MC}/\text{eV}) = 18.76^{+0.03}_{-0.06} \text{ (MC sim. threshold)}$$

$$\lg(E_{thres}/\text{eV}) = 18.5 \pm 0.1 \text{ (Hybrid threshold)}$$

The value and uncertainty of E_{MC} is taken from the calibration fit with Aachen muon map. The uncertainty of E_{thres} is a conservative estimate from Fig. 14. Even with a corrected energy scale, the numbers are not compatible. A more detailed analysis on this observation will be postponed for now.

By studying Fig. 15 and by varying the energy threshold, we can draw the following conclusions.

- The γ -values of Aachen, Santiago and Paris are compatible within 1σ and also with their equivalent in vertical event analysis: $\gamma_{vert} = 1.05 \pm 0.02$ [25]. The value of Paris however carries huge uncertainties because of the low statistics.
- The large bulk of calibration events comes from the angular region $\theta \in [60^\circ, 70^\circ]$. Thus, a reliable reconstruction of the data below 70° is imperative for the energy calibration. To extrapolate this calibration correctly to higher zenith angles, it is important that the attenuation is correctly described by the muon map.
- The calibration constants are stable within their uncertainties for small ($\lg E_{thres}/\text{eV} = 18.5 \pm 0.1$) variations of the energy threshold.
- Events below the energy threshold show less scattering around the calibration line in the data set reconstructed with the Aachen map, than in the data set of the Santiago map. This confirms previous observations, that the Aachen muon map more reliably fits events at low energies.
- For events above the energy threshold, the Santiago and Aachen muon map fit the data comparably well. The observed difference in the energy resolution is not significant within current statistics. Together with the last point this shows, that for events with a high tank multiplicity, the reconstructed value of N_{19} becomes rather insensitive to the shape of the muon map.

The data set used for the calibration currently is very small, but expected to increase rapidly as the

observatory is now nearly complete. The successful energy calibration for every map allows further muon map comparisons.

4.3.3 Energy comparison

After calibrating the SD data with hybrids for each map type, we can analyse the average energy shift around several benchmark points in the θ, ϕ -plane (see Fig. 17).

The largest shift can be observed between Paris and the other maps, which is about 20 %. Aachen and Santiago show a small total difference of 7 %, but the energy shift varies strongly with zenith angle. We observe 20 % at $\theta \simeq 62^\circ$, 0 % at $\theta \simeq 72^\circ$ and roughly 10 % again at $\theta \simeq 79^\circ$. The evolution of the energy difference with θ roughly matches with the evolution of the amplitude difference in Fig. 12. This confirms the definition of the amplitude in Section 20, p. 15.

It can be checked, that the large energy shift in the Paris data relative to the other ones is caused by the poor energy calibration fit. The Paris map data can be brought to into agreement with the Aachen map data, if the calibration constants from the Aachen muon map are used.

We conclude, that the average energy shift caused by choosing one muon map over the other is at the 7 % level. However, changing the muon map causes fluctuations of the reconstructed energy up to 20 %. The large shift between the Paris map and the other maps is caused by a poor calibration and is not a feature of the map itself.

4.3.4 $\sin^2(\theta)$ distribution

Since we observed energy shifts between the Santiago and Aachen muon map related to the overall map attenuation with zenith angle, it is reasonable to further investigate the $\sin^2(\theta)$ distribution of each reconstructed data set. An isotropic primary flux hitting a flat horizontal detector essentially leads to a flat $\sin^2(\theta)$ distribution. This can be shown by asking how many rays go through two infinitesimal areas, one perpendicular to the isotropic flux dA_o and another one horizontal to

the ground dA_g :

$$\begin{aligned} \frac{dA_o}{dA_g} &= \cos \theta \\ 1 &\propto \frac{dN}{dA_o} \text{ (Isotropy)} \\ &= \frac{dN}{dA_g} \frac{1}{\cos \theta} \\ &\propto \frac{dN}{d \cos \theta} \frac{1}{\cos \theta} \propto \frac{dN}{d \sin^2 \theta} \end{aligned} \tag{29}$$

We used here, that the area dA_g is exposed to the solid angle $d\Omega = d \cos \theta \cdot d\phi$ of the source.

In Fig. 18, we count all events above the threshold energy $\lg(E_{\text{thres}}/\text{eV}) = 18.5$ for each muon map. If the attenuation of the muon map fits the data, the distributions needs to be flat. We can see, that this is roughly correct for all maps, but there are also deviations from uniformity.

The Santiago map shows a constant increase of the estimated energy with growing zenith angle, which is consistent with the previously observed shifts in map amplitudes and reconstructed energy compared to the Aachen map. The Aachen map shows some fluctuations, but there is no overall upward slope. The largest deviation can be found at $\theta = 77^\circ$, which would indicate an underestimation of the muon density at this inclination. The Paris map also shows a deviation from uniformity near 70° and near 80° . This is mainly due to bad calibration. If the Aachen map calibration is applied to the Paris map, the flatness improves a lot.

We repeated the $\sin^2(\theta)$ analysis with a higher cut energy of $\lg E/\text{eV} = 19.0$. The fits improve, in particular the Aachen muon map now agrees perfectly the attenuation found in data.

Muon map	χ^2/ndof
Santiago	1.4
Paris	2.0
Aachen	1.0
Paris (AC cal.)	2.2

Table 5: Agreement of the $\sin^2(\theta)$ distribution found in data with a uniform distribution, for events with $\lg(E/\text{eV}) > 19.0$. The number of events per bin in this fit is around 50.

We conclude, that the Aachen muon map currently shows the best match of muon attenuation

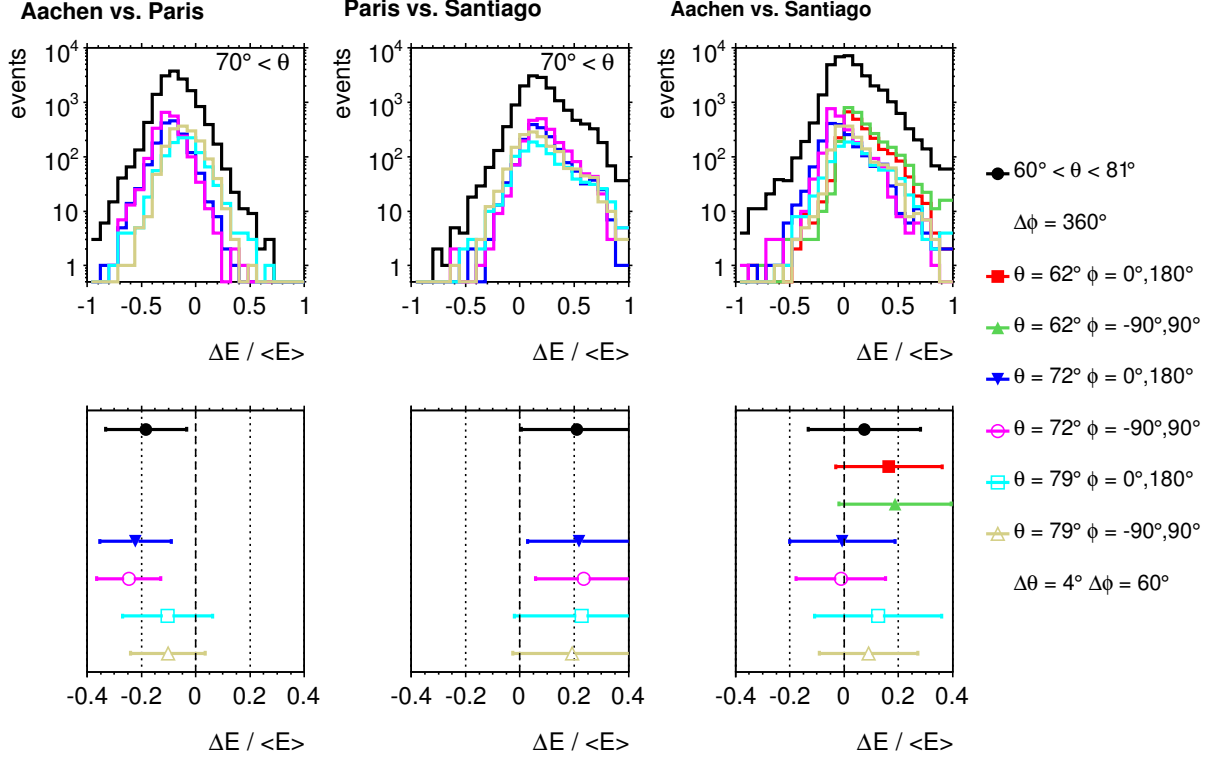


Figure 17: Relative change in the reconstructed calibrated energy by choosing a different muon map. The bottom row shows the mean and through the bars the RMS of the distributions from above. Collected are the values obtained at different "benchmark regions" in the θ, ϕ -plane. The legend shows the mean value and angular size of each region. For the Paris map, values below $\theta = 70^\circ$ are not available.

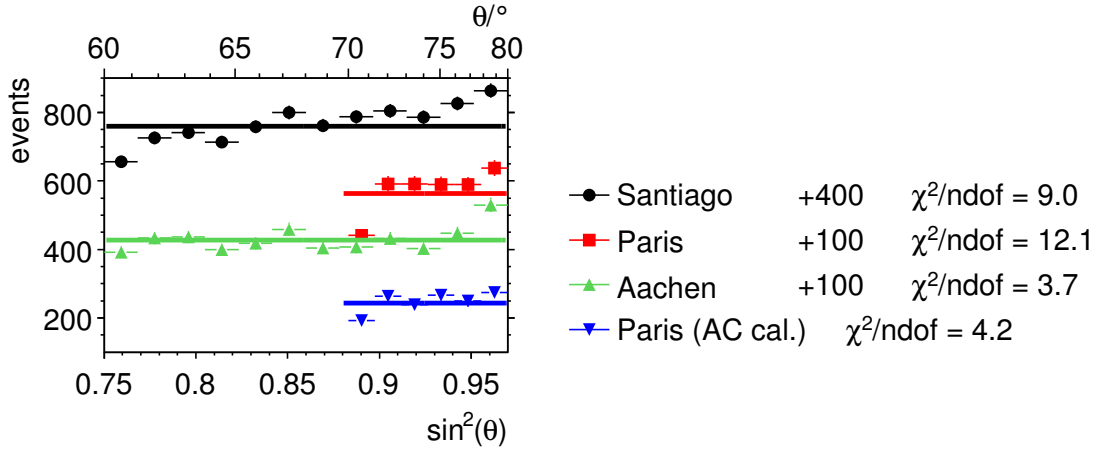


Figure 18: $\sin^2(\theta)$ distribution for T5 selected SD events above $\lg(E/\text{eV}) = 18.5$, reconstructed with different muon maps. To easier distinguish the histograms, they are shifted by the constant given in the legend. Each distribution was fitted with a constant line to show the agreement with the theoretical expectation.

with rising zenith angle with data. A wrong description of attenuation in the muon map decreases the accuracy of the calibration fit and therefore the energy resolution.

4.3.5 Comparison of energy spectra

Using the reconstructed and calibrated data sets for all muon maps and the calculated exposure in the regarded time frame, we can derive three energy spectra, one for each muon map (see Fig. 19). We only show points above $\lg E/\text{eV} = 18.4$, because of the SD trigger saturation threshold derived in Section 4.3.2, p. 22.

For the Aachen map, the flux uncertainty caused by the uncertainty of the hybrid energy calibration and in N_{19} was numerically propagated into the plot by calculating 500 realisations of the energy spectrum. In each realisation, the calibration constants and each N_{19} are fluctuated once within their uncertainties assuming a gaussian distribution. In case of calibration constants, also the correlation found in the fit was taken into account. The resulting spread in each bin is shown in the plot.

The Santiago and Aachen map spectra are in perfect agreement with each other and with the vertical spectrum. The Paris spectrum disagrees with the other spectra below 10^{19} eV, but this is again an effect of the poor calibration. Perfect agreement can be achieved, if the calibration of the Aachen map is used also for Paris data set.

We conclude, that the energy spectrum is insensitive to the choice of the three muon maps. The two main reasons for this insensitivity are the energy calibration with FD hybrids and the large multiplicity found in inclined events above $10^{18.4}$ eV. Both make sure, that the average reconstructed energy is correct. On the other hand, a wrong attenuation or shape in a muon map decreases the energy resolution and increases the uncertainty of the calibration constants. But since the current statistics are limited, especially in the calibration, no observable difference can be found in the spectrum.

5 Conclusion

We presented a new muon map, based on a CORSIKA/QGSJet-II/FLUKA proton shower library. The new map provides a continuous descrip-

tion of the muon ground profile for zenith angles $\theta \in [60^\circ, 80^\circ]$. It includes Early-late asymmetries, that mainly result from the curvature of the shower front. The method described in this note to generate the muon map can be used with practically any other general purpose shower library, because it makes no restrictions on the distribution of showers in the (E, θ, ϕ) parameter space.

An extensive comparison was done between muon maps of the IPN Paris group [11] and the USC Santiago de Compostela group [10] and the one presented in this note. Shape and attenuation with zenith angle differ significantly in all candidates. Combining several tests, it was found that the Paris and Aachen map describe the real signal profile slightly better than the Santiago map. The Aachen muon map turned out to be the best model of the signal attenuation with zenith angle found in data.

Using SD and FD hybrid data, a calibration was performed for each map with an energy cut derived also from hybrid data. Despite the differences in the muon maps, only negligible systematic energy shifts of 7 % were found after the calibration. However, random energy shifts at the 20 % level were observed.

The inclined energy spectrum was derived from SD data after the reconstruction with each map. The spectrum was found to be insensitive to the choice of the muon map within the current statistical and systematic uncertainties.

The analysis also showed, that a good reconstruction of showers in the overlap region between vertical and inclined showers $\theta \in [60^\circ, 70^\circ]$ is imperative for the hybrid energy calibration. This implies a good understanding of the primary electromagnetic signal, which gives a varying and non-negligible contribution to the total signal in this region.

Further, we found that the signal uncertainty is largely underestimated in the currently used reconstructions, because the fluctuation of the electromagnetic signal component is neglected.

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We like to thank Enrique Zas for providing the shower reconstruction code of the USC Santiago, without which large parts of this analysis would

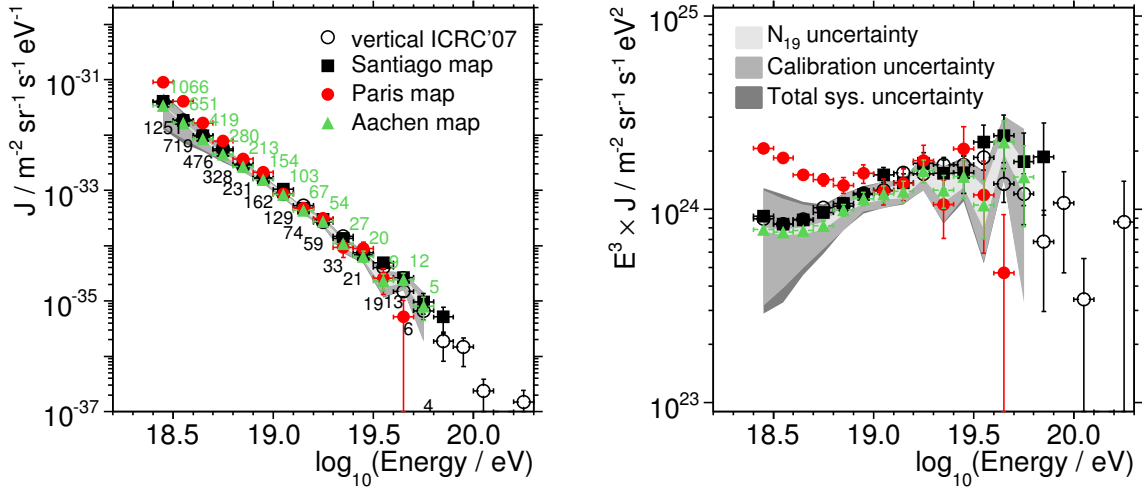


Figure 19: Energy spectrum of inclined showers for $\theta \in [60^\circ, 80^\circ]$ between 2004/01 and 2007/03, from three data sets, each reconstructed and calibrated with one of the muon maps of Santiago, Paris and Aachen. For comparison, the standard vertical spectrum published in [26] is underlaid. Only statistical uncertainties are included for the Santiago and Paris map spectra. For the Aachen map spectrum, the flux uncertainty caused by the energy uncertainty was numerically evaluated (see text) and is shown by the colored areas: 1) light grey - propagated uncertainty of N_{19} , 2) middle grey - propagated uncertainty of the calibration constants, 3) dark grey - combined uncertainty.

not have been possible. We further like to thank Olivier Deligny for his help with the Paris muon map, and acknowledge Veronique van Elewyck, Pedro Facal, Víctor Olmos-Gilbaja, Inés Valiño and Ricardo Vázquez for their valuable correspondence and insights on various aspects in this analysis.

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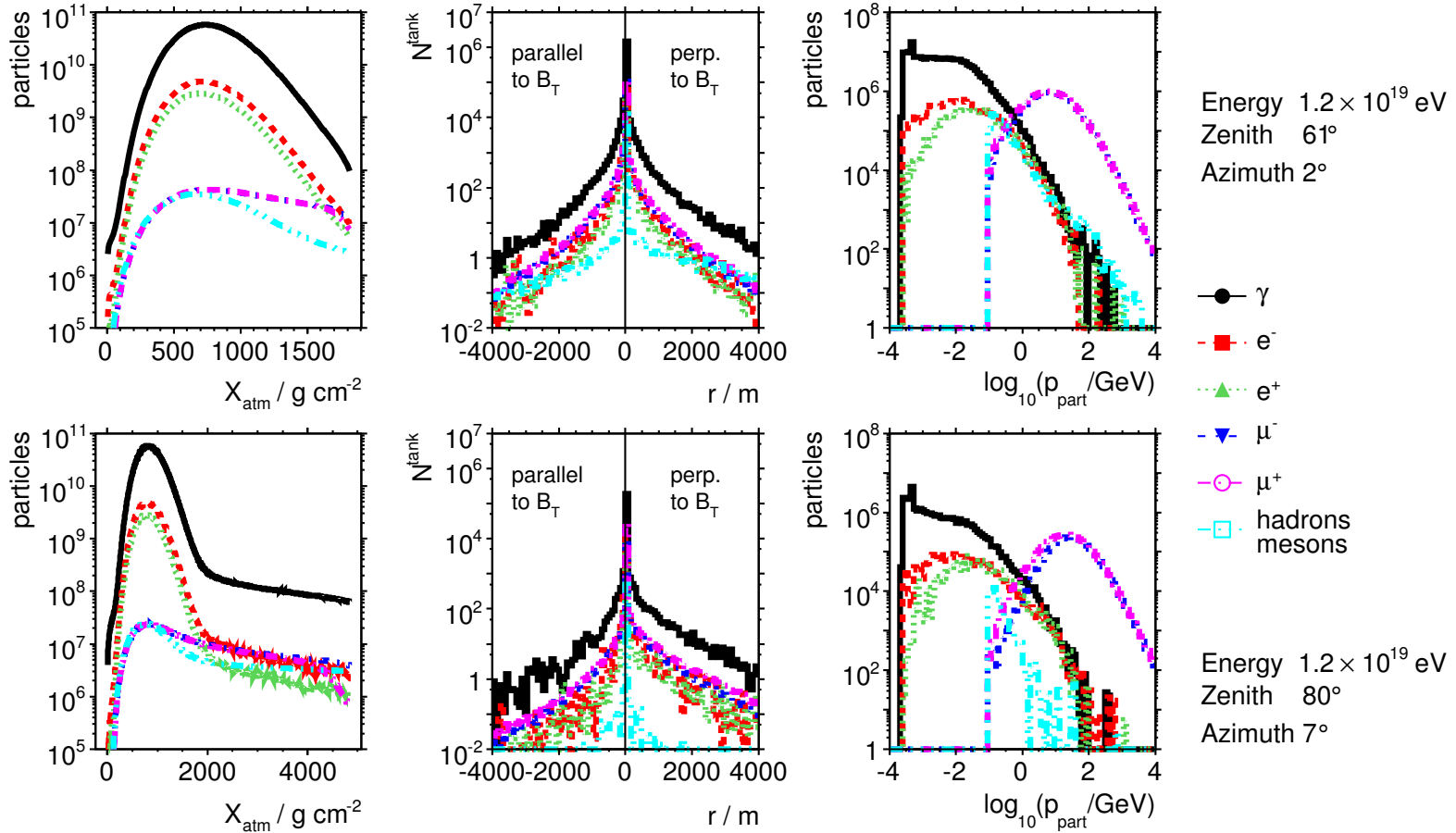


Figure 20: Comparison of two inclined proton showers, simulated with CORSIKA/EPOS/FLUKA [4, 6, 14]. From left to right: 1) total particle number vs. passed atmospheric depth, 2) particle number per tank on the ground vs. radial distance from the shower axis, 3) energy distribution on the ground. In 2), $r > 0$ relates to particles collected within $\pm 10^\circ$ around the direction perpendicular to the projected geomagnetic field B_T and $r < 0$ relates to particles collected within $\pm 10^\circ$ around the direction parallel to B_T (see Section 2, p. 4). In all plots, only particles above certain energy thresholds are shown: $E_{\text{thres}} = 250$ keV for electromagnetic particles and $E_{\text{thres}} = 0.1$ GeV for everything else. The low energy photon peak in 3) is caused by e^+/e^- annihilation. The particle depletion along B_T in 2) for $\theta = 80^\circ$ is the result of geomagnetic deflection.

$\lg(E/\text{eV}) = 19.00$ Zenith = 79.2° Azimuth = -0.4°

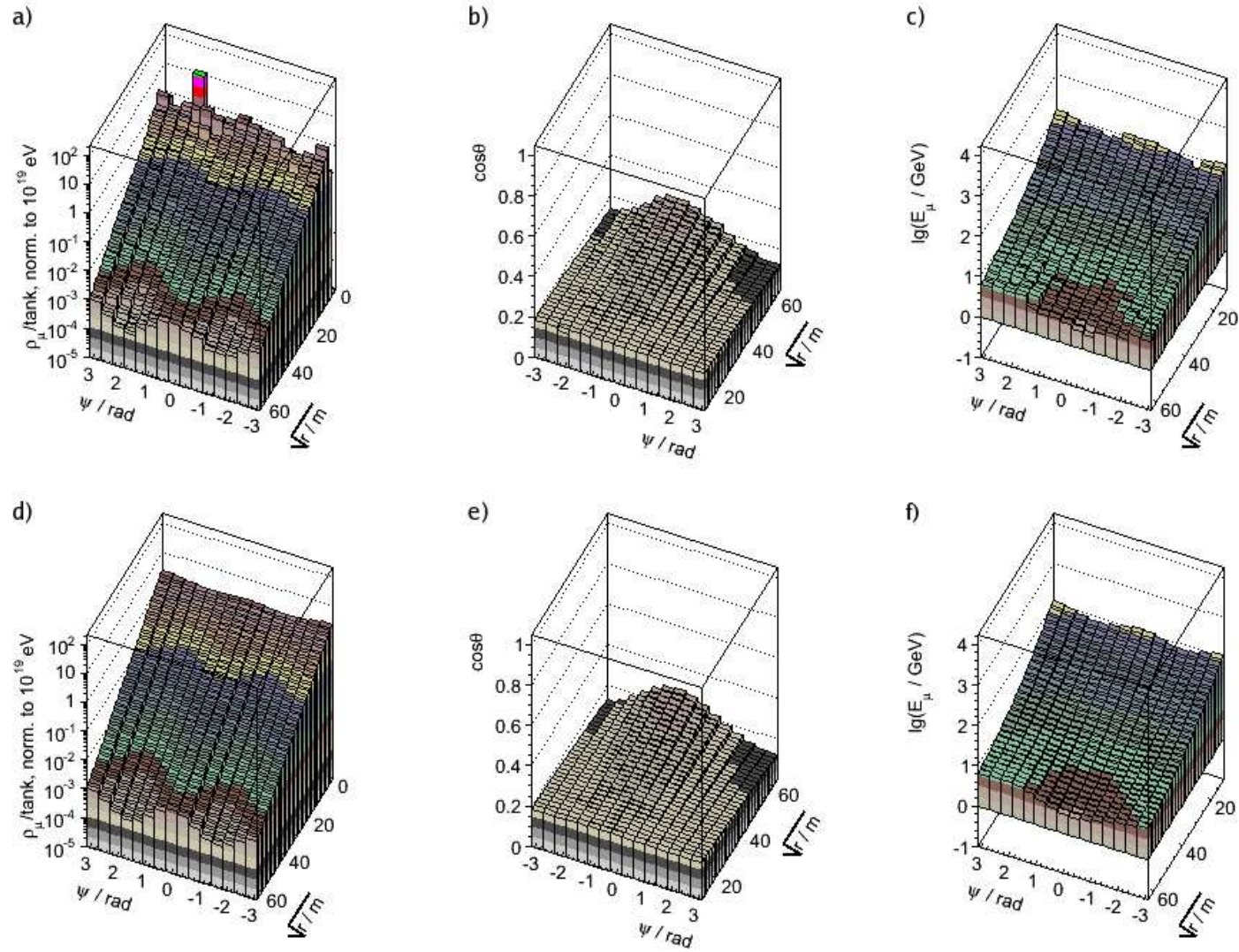


Figure 21: Example fit of the lateral muon distribution on the shower level. Plotted are the distributions of a) the number of muons per tank, b) the cosine of the muon inclination and c) the muon energy, as they are obtained from the simulation. d), e), f) show the distributions after the fit.

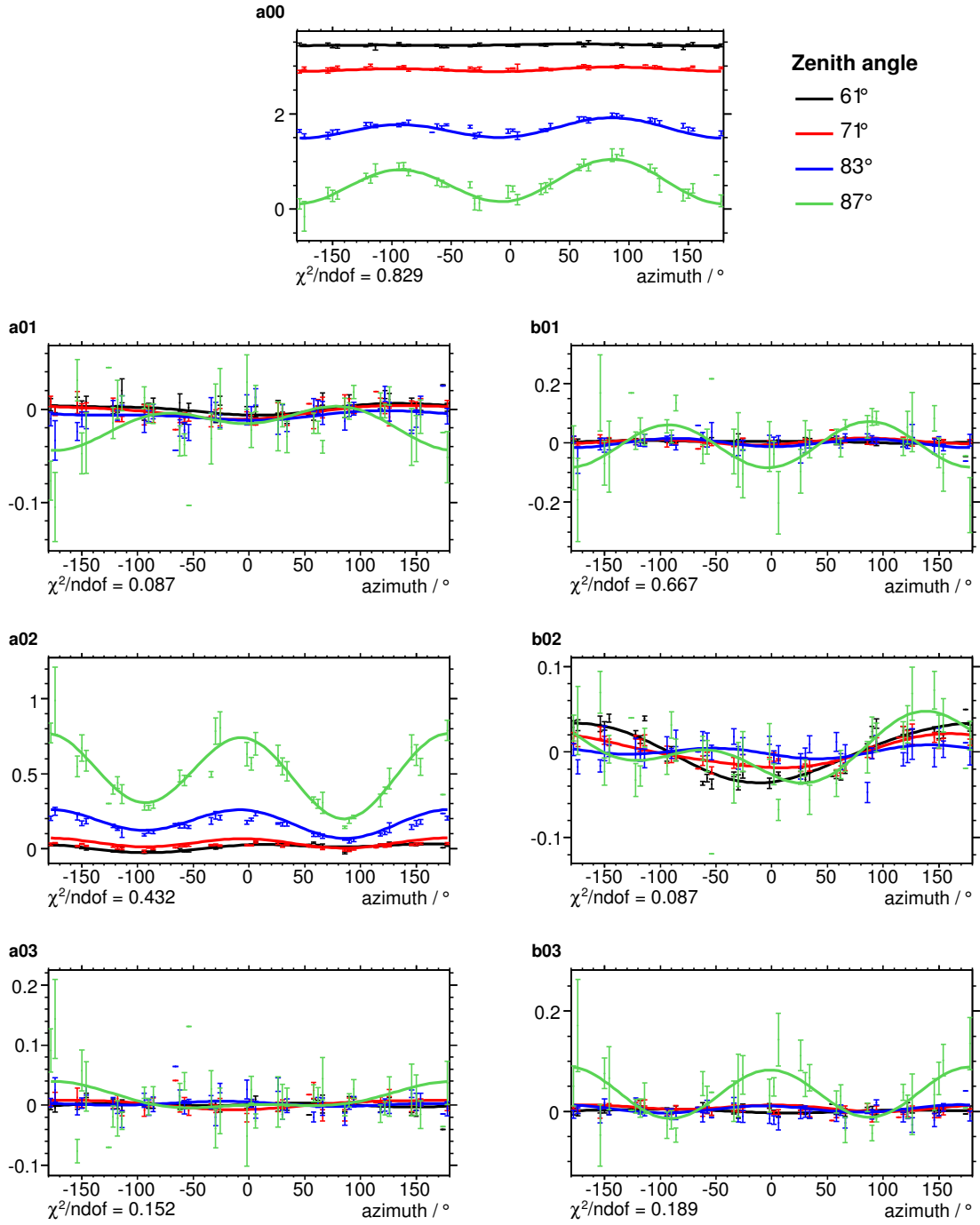


Figure 22: Second level fits to map parameters of the lowest order. The fitted function is a continuous 2-dim. shape. The plots shows projections of this shape at a set of zenith angles (see the legend). Overlaid is a profile of the data near these zenith angles.

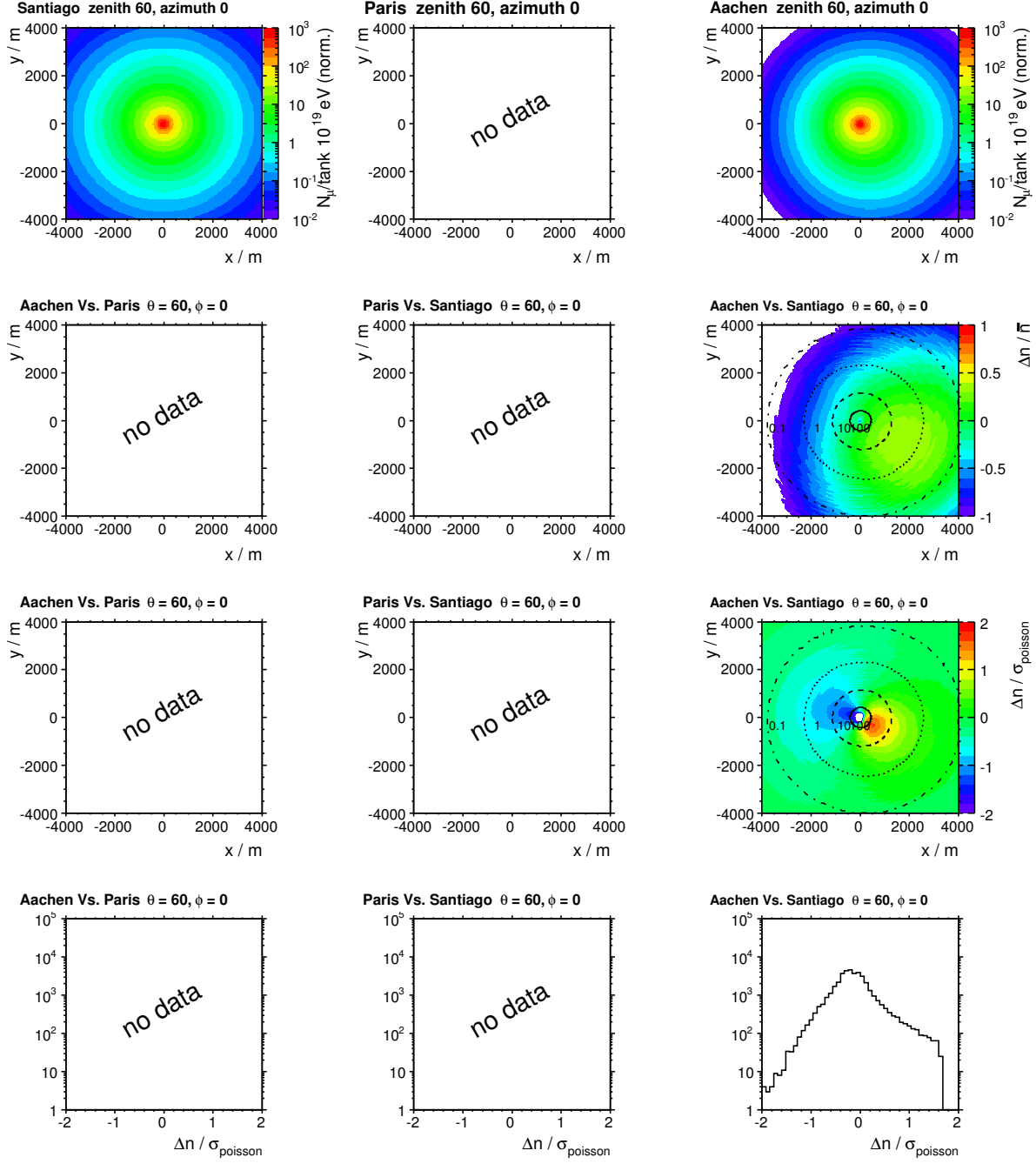


Figure 23: Shape comparison of different muon map types. The amplitude N (see Eq.20) in all maps was normalised to the Santiago map. First row: raw map. Second row: pair-wise relative difference. Third row: pair-wise difference divided by the local Poisson uncertainty. The forth row: histograms of the above distribution. Below $\theta = 70^\circ$, the Paris map is not defined (see the following figures).

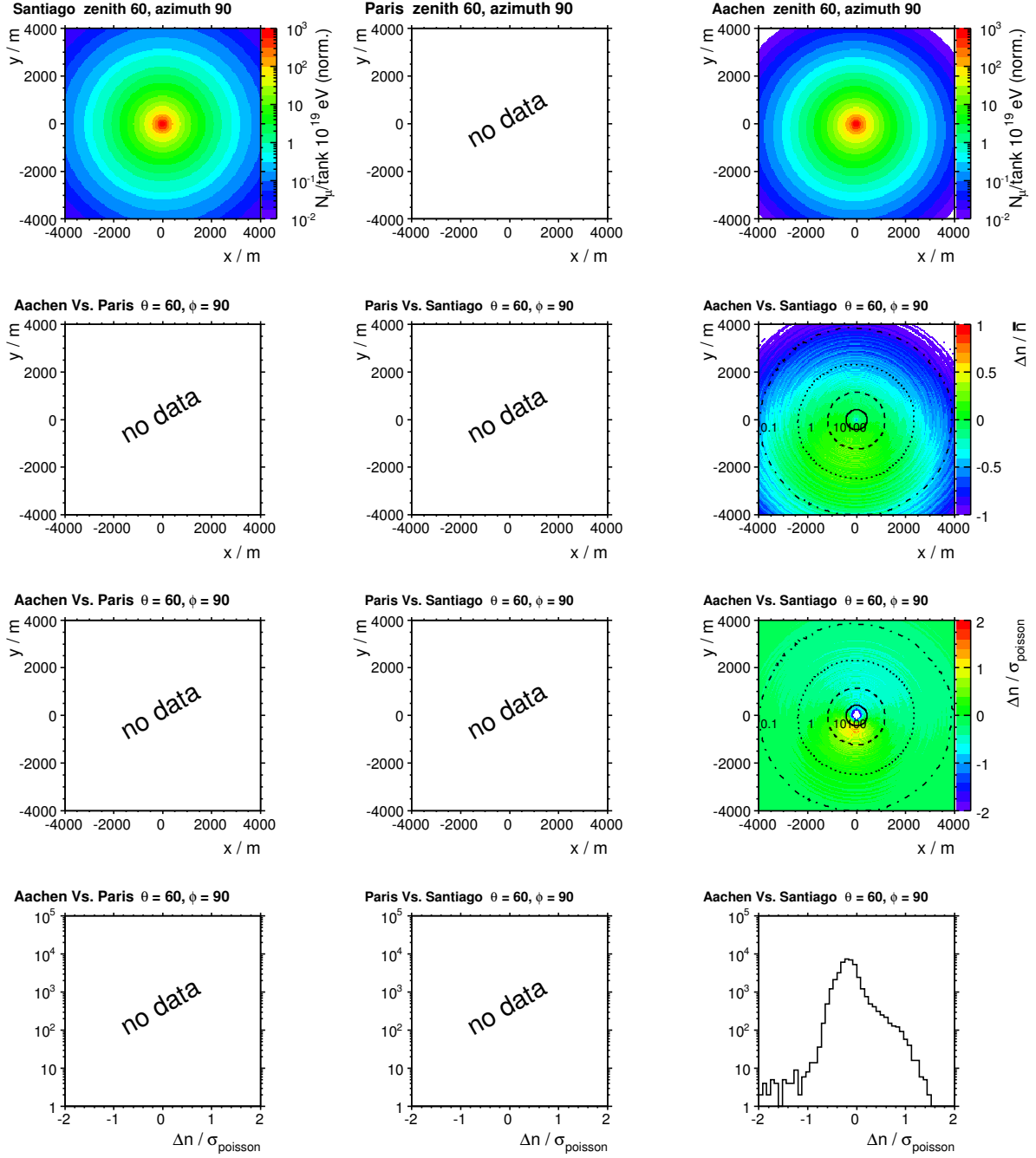


Figure 24: Comparison of the shape of different muon maps. See Fig. 23.

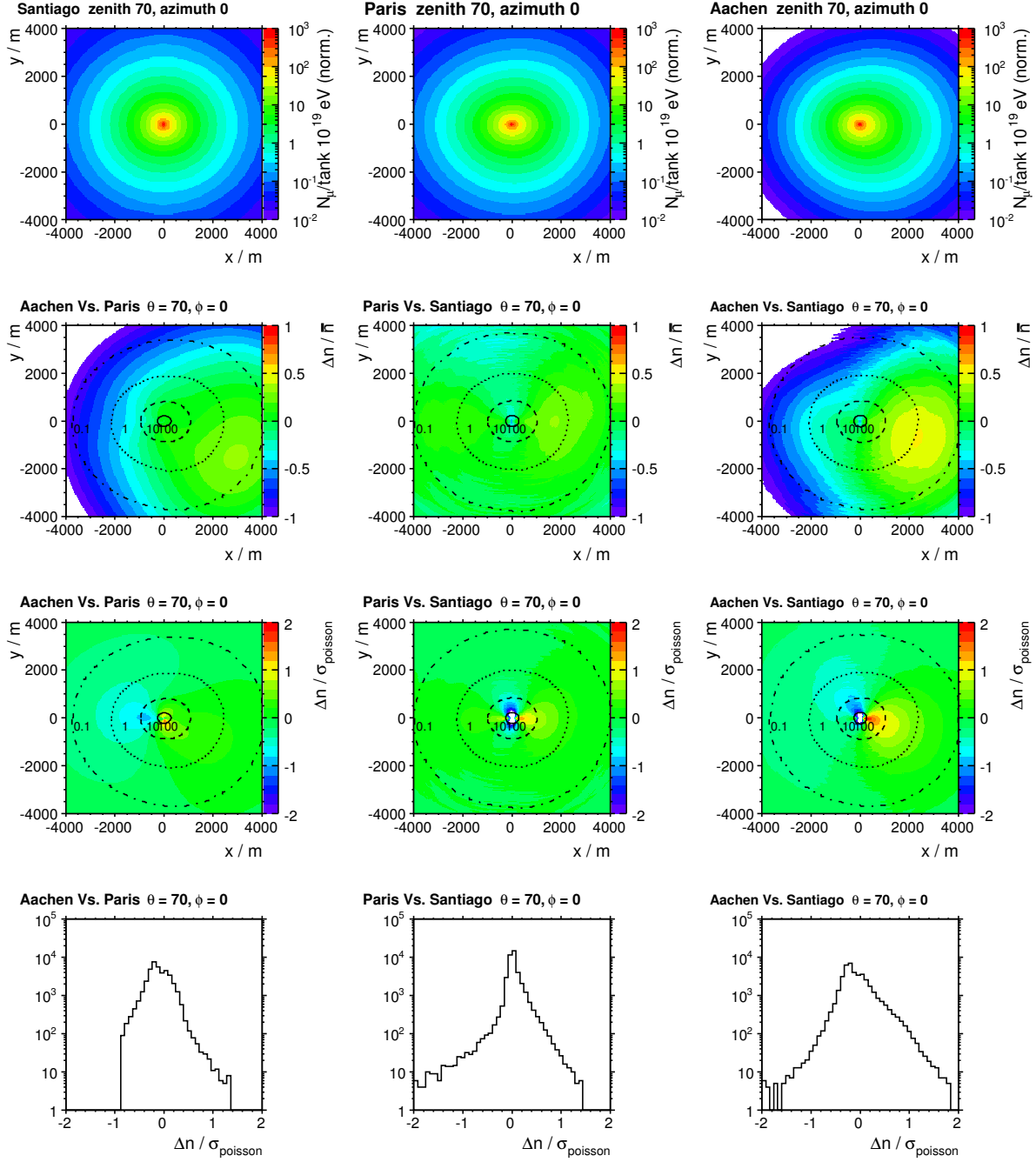


Figure 25: Comparison of the shape of different muon maps. See Fig. 23.

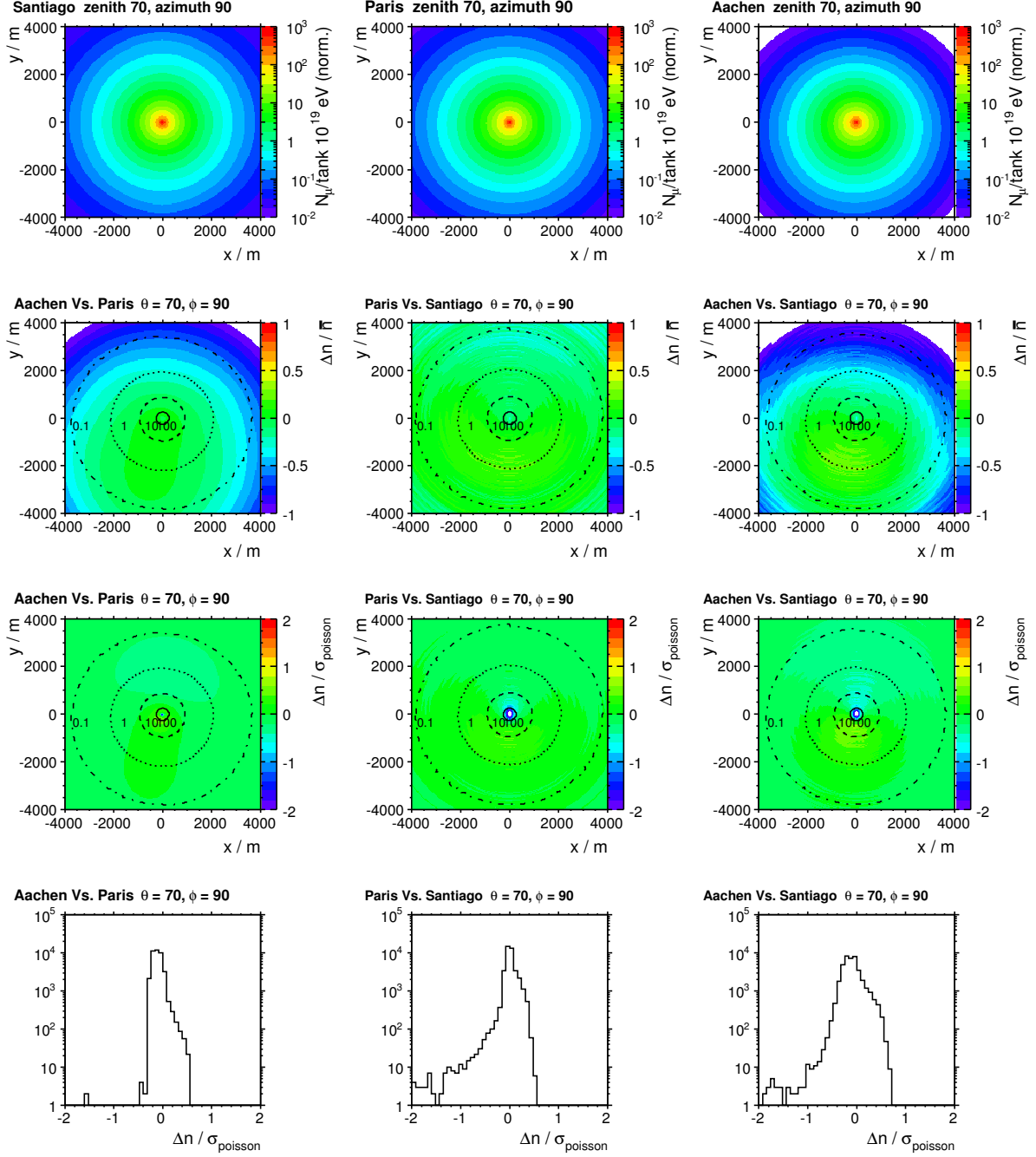


Figure 26: Comparison of the shape of different muon maps. See Fig. 23.

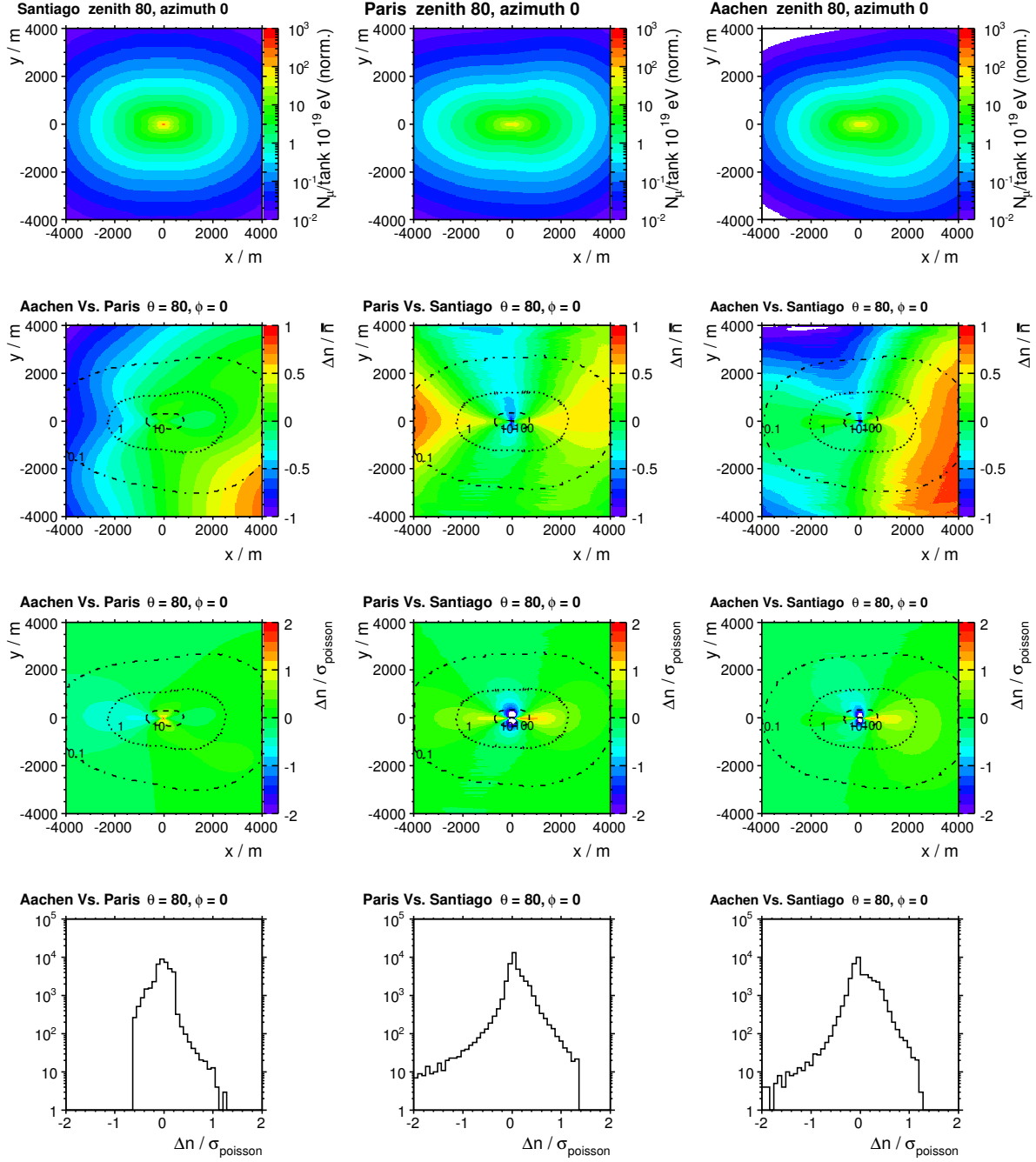


Figure 27: Comparison of the shape of different muon maps. See Fig. 23.

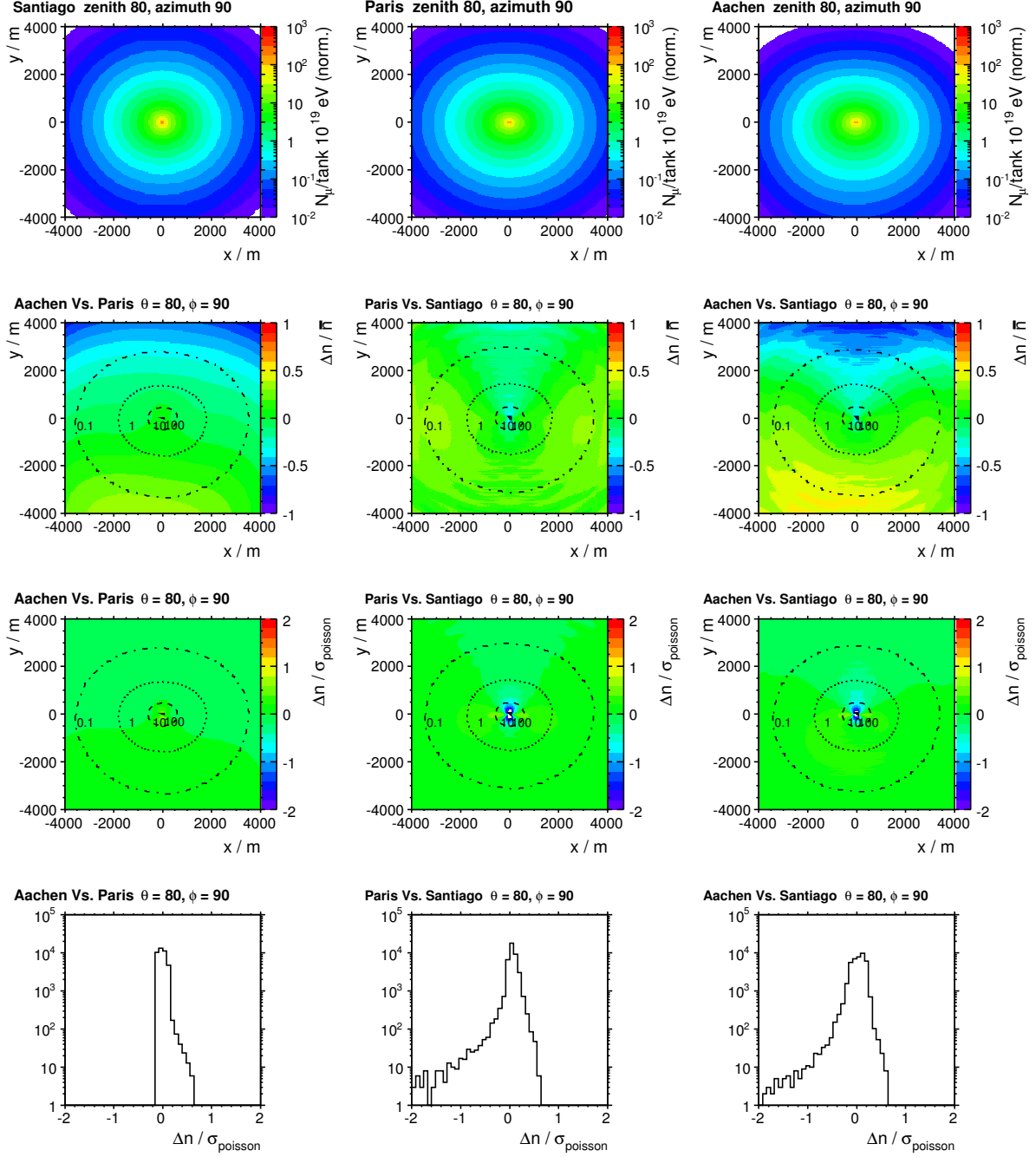


Figure 28: Comparison of the shape of different muon maps. See Fig. 23.

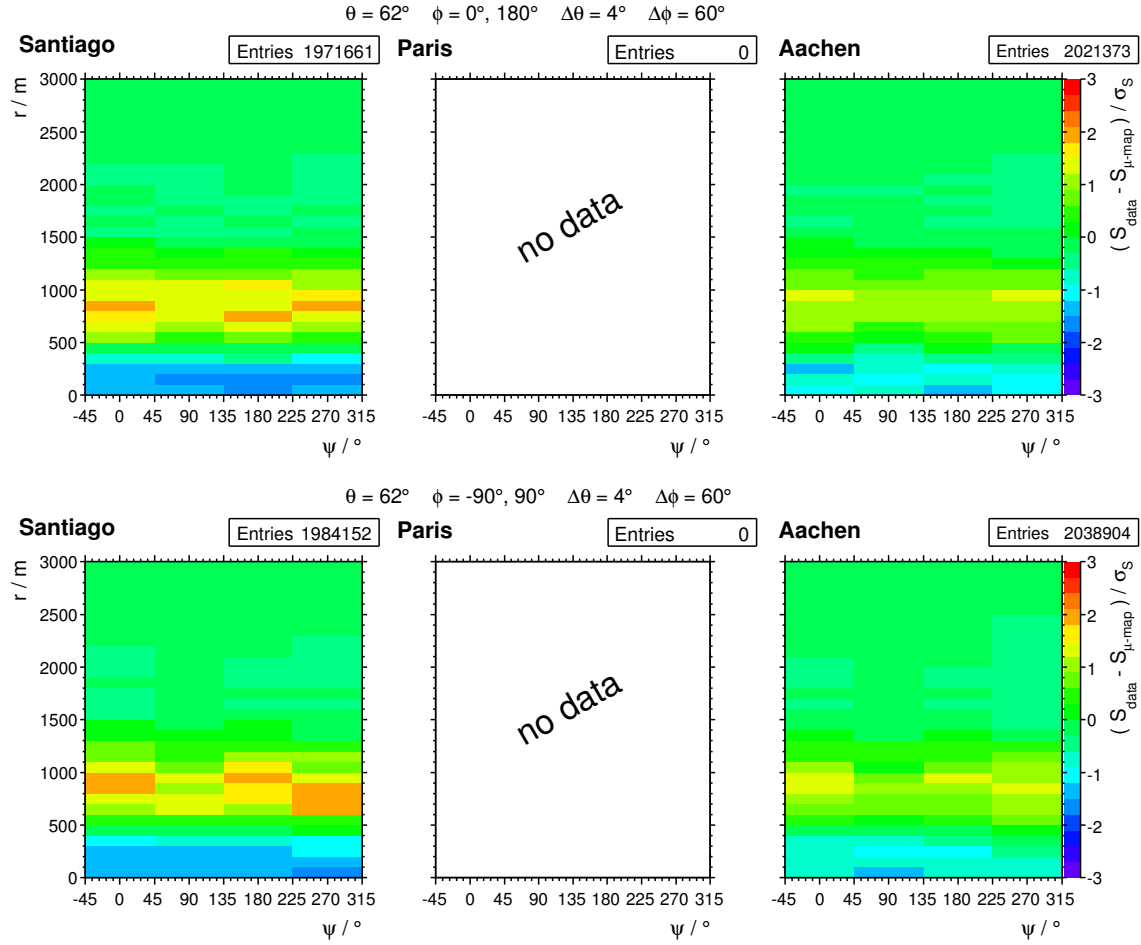


Figure 29: Average difference between measured and expected signal divided by the signal uncertainty binned in the shower plane. This profile only includes events between $\theta = 60^\circ$ and 64° .

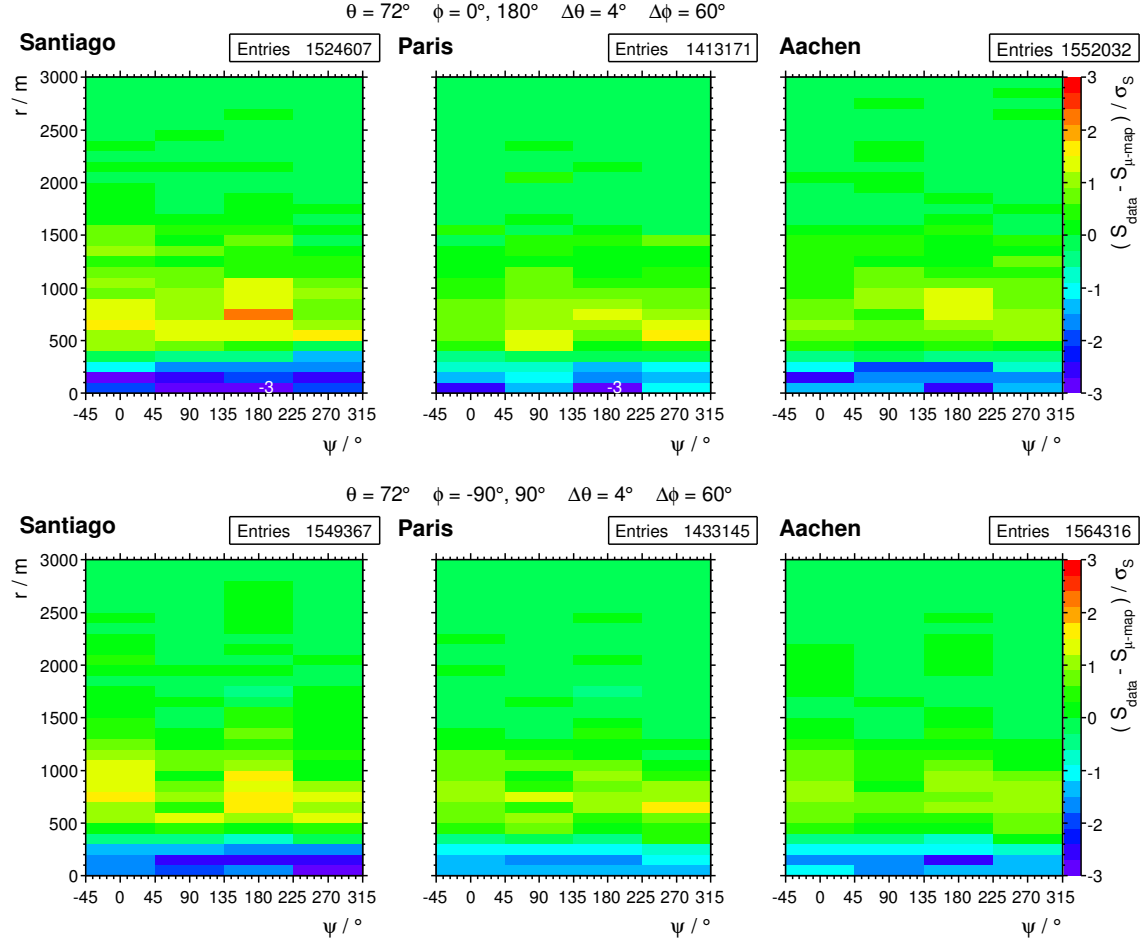


Figure 30: Average difference between measured and expected signal divided by the signal uncertainty binned in the shower plane. This profile only includes events between $\theta = 70^\circ$ and 74° .

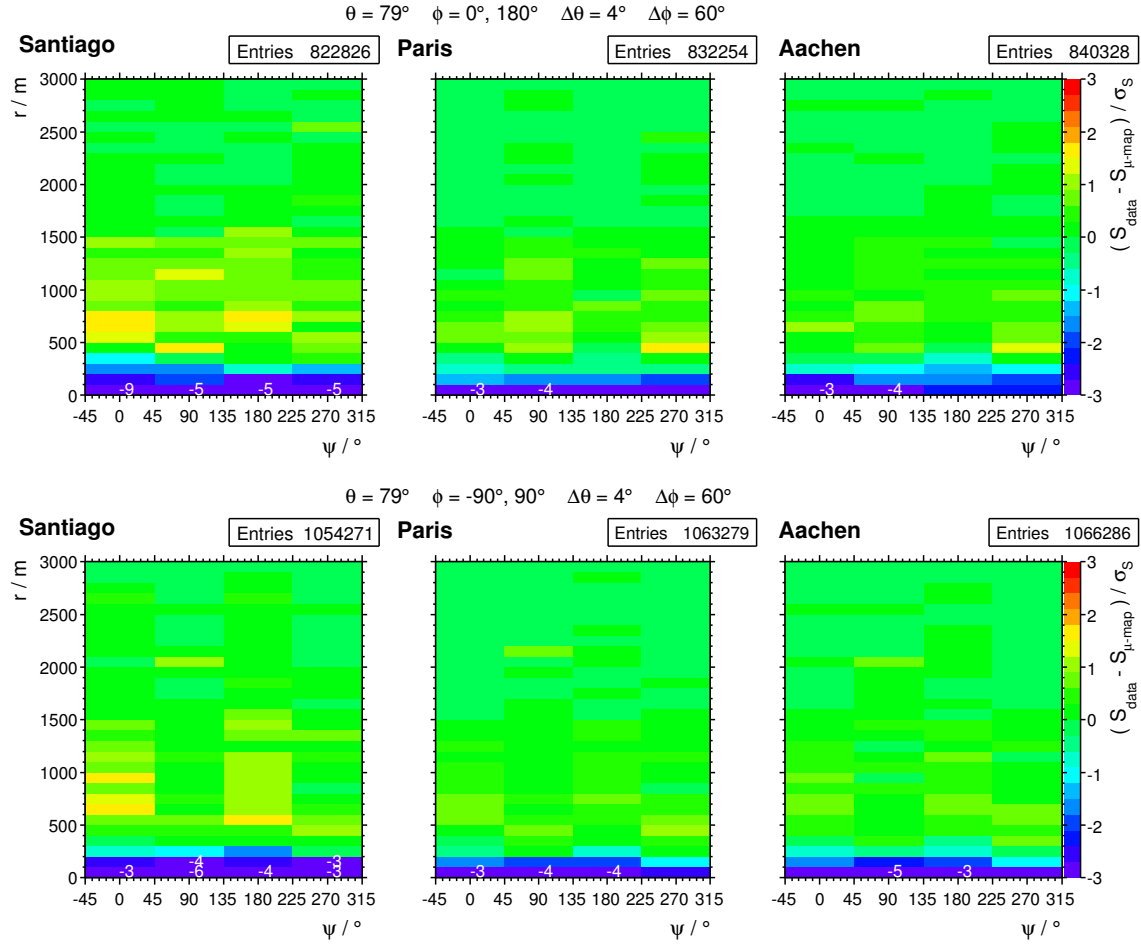


Figure 31: Average difference between measured and expected signal divided by the signal uncertainty binned in the shower plane. This profile only includes events between $\theta = 77^\circ$ and 81° .